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Revisiting the Liantuo Formation in Yangtze Block, South China: SIMS U–Pb zircon age constraints and regional and global significance

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ABSTRACT

The age and stratigraphic correlation of the Liantuo Formation in the Yangtze Block have been controversial. Disputes diverge on whether it is correlative with the Xieshuihe Formation, the Fulu Formation, the Chang'an Formation or the Banxi Group. Some radiometric dates have been obtained from the Liantuo Formation but they suffer from inheritance. In this paper, we carried out a detailed SIMS U–Pb zircon examination of the Liantuo Formation. Radiometric dates from six tuff/tuffaceous siltstone beds within the Liantuo Formation suggest it was deposited in a time interval of ca. 780–714 Ma, indicating its correlation with the Banxi Group. New ages not only coordinated the age-paleopole contradictions in the Liantuo and Nantuo formations for exploring the evolution history of Yangtze Block during the breakup of Rodinia supercontinent, but also strengthened the viewpoint that the Sturtian glaciation occurred at ca. 715 Ma in at least three paleocontinents of Arabia-Nubia, Laurentia, and South China. Some zircon grains in samples YC20 and 13YCB develop bright domains of ca. 690–680 Ma. These domains have lower total REE, lower trace element contents, and lower Th/U ratios than cores. However, both the bright domains and cores show similar $\delta^{18}O$ values lower than the normal mantle zircon values. These characteristics suggest that these domains were possibly formed by post-depositional recrystallization.

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1. Introduction

The Nanhua rift basin in the South China Block (SCB) accommodated a wealthy of well-preserved Neoproterozoic (850–635 Ma) volcano-sedimentary records (Wang and Li, 2003; Li et al., 2008a,c). Such a suite of intermingled sedimentary and volcanic rocks is well outcropped in the Hubei–Hunan–Guizhou–Guangxi regions hosting vital information linking to the evolution history of Neoproterozoic Rodinia supercontinent (Li et al., 2008c) and Neoproterozoic 'snowball Earth' event (Zhang et al., 2008a; Hoffman and Li, 2009; Macdonald et al., 2010). The Liantuo Formation, as the first sedimentary cover on the Yangtze Craton around the Three Gorge area, Hubei Province, is just one example of such stratigraphic units whose sedimentation have potentially bridged the evolution of Yangtze Block during Rodinia breakup and initiation of the Jiangkou glaciation (equivalent of Sturtian glaciation, Zhang and Chu, 2006; Zhang et al., 2008a).

Deposition of the Liantuo Formation is closely associated with evolution history of Rodinia supercontinent as it witnessed the breakup of Yangtze Craton in South China. Previous palaeomagnetic data allows for placing the Liantuo Formation at an average palaeolatitude of 33° N at ca. 750 Ma (Evans et al., 2000; Li et al., 2013b), which is within errors concordant with the average palaeolatitude of 35° N from the Nantuo Formation at ca. 650 Ma (Zhang and Piper, 1997; Zhang et al., 2013). These identical paleopoles, however, are contradictive with the discrepant radiometric ages as it is hard to explain why the paleopoles remained unchanged within 100 myrs time interval (Li et al., 2013b; Zhang et al., 2011, 2013, 2014). Feasible solutions are either assigning a younger palaeomagnetic pole to the Liantuo Formation (Li et al., 2013a,b) or assuming a loop existed in the apparent polar wander path from ca. 750–650 Ma (Zhang et al., 2011). The former solution can be more promising as sporadic ca. 730–700 Ma zircon U–Pb ages have been reported from the upper part of Liantuo Formation (Liu et al., 2008; Gao and Zhang, 2009), indicating its depositional age could be much younger than previously thought, although detection of younger zircon U–Pb ages are required.

Redefining the depositional ages of Liantuo Formation also contribute to resolving the disputes on regional stratigraphic

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correlation. Lack of convincing radiometric dates has confused Chinese geologists more than three decades with regard to the stratigraphic correlation of Liantuo Formation, with possible attributions to the Xieshuihe Formation (Zhang, 1986a,b; Wang and Suo, 1987; Liu and Zhang, 1989; Huang et al., 1996a,b; Xue et al., 2001; Peng et al., 2004; Wang et al., 2006a,b), Fulu Formation (Lin, 1995; Xue et al., 2001; Peng et al., 2004; Lin et al., 2013), Chang'an Formation (Lu et al., 1985; Wang et al., 2006a,b, 2009; Yin and Gao, 2013; Gao et al., 2013a) or Banxi Group (Liu et al., 1991; Liu and Li, 1992; Wang and Li, 2003; Zhang et al., 2003, 2008b, 2011; Feng et al., 2006; Zhang and Chu, 2006; Du et al., 2013) from different perspectives. A precise and accurate chronological solution is required in order to place the Liantuo Formation in a suitable position in the stratigraphic framework of the Nanhua Basin. The aim of this paper is to provide an accurate radiometric age control on the Liantuo Formation by means of dating zircon grains from interbedded tuff/tuffaceous siltstone beds. In doing so, its correlation with other stratigraphic units in the Nanhua Basin and its connection with the evolution of Rodinia supercontinent and Jiangkou/Sturtian glaciation will be clarified.

2. Geological setting and sampling details

2.1. Geological setting

Amalgamation of the Yangtze Block to northwest and the Cathaysia Block to southeast along the Jiangnan Orogen led to the formation of South China Block (SCB) during the late Mesoproterozoic to earliest Neoproterozoic Jiangnan Orogeny (Fig. 1a; Li et al., 2007, 2008b, 2009a). Subsequently, the SCB experienced intracontinental rifting, and accommodated a wealthy of well-preserved Cryogenian sedimentary and volcanic rocks in the Nanhua Basin – the largest Neoproterozoic basin in SCB (Wang and Li, 2003; Li et al., 2008a, 2008c). The sedimentary rocks are well outcropped in Hubei–Hunan–Guizhou–Guangxi regions of the Nanhua Basin (Fig. 1a), with a depositional environment transitioned gradually from onshore in northeast to deep marine in southwest. Total thickness of the Cryogenian sedimentary rocks varies from ca. 300 m in northeast to >3000 m in southwest.

Updated geochronological and geochemical data combined with differential lithological associations suggest that the Cryogenian successions are dividable into three tectonostratigraphic sequences (Wang and Li, 2003; Wang et al., 2012a and references therein; Lan et al., 2014). They are, in ascending order, the Sibao Group (ca. 850–820 Ma) in Guangxi and Guizhou provinces (equivalent of Lengjiaxi Group in Hunan Province and Shuangqiaoshan Group in Jiangxi Province) as the first sequence, the Banxi Group (ca. 814–715 Ma) in Hunan Province (equivalent of Danzhou Group in northern Guangxi Province and Xiajiang Group in eastern Guizhou Province) as the second sequence, and the Jiangkou Group, Datangpo and Nantuo formations (ca. 715–635 Ma) as the third sequence characterized by glacial and interglacial deposits (Fig. 2). The Jiangkou glaciogenic deposits were divided into Chang'an Formation and Fulu Formation which includes the above stadial Gucheng Member and below interstadial Liangjiehe/Xieshuihe Member (Zhang and Chu, 2006). The Gucheng Member correlates with Tiesi'ao and Dongshanfeng members on the grounds of common presence of glacial diamictite. Likewise, the Liangjiehe Member has been correlated with Xieshuihe Member based on their similar lithological features, they make up the Fulu Formation together with Gucheng/Tiesiao/Dongshanfeng Member (Zhang and Chu, 2006). The Gucheng Member is overlain conformably by the Datangpo Formation, which also conformably underlies the Nantuo Formation.

Unlike the Nantuo Formation that is widely exposed in the Nanhua Rift Basin, outcrops of the Liantuo Formation are mostly confined to northern and western Yichang City, Hubei Province. The Liantuo Formation was initially termed by Lee (1924) as the Nantuo grit, and subsequently was re-assigned by Liu and Sha (1963) as Liantuo Formation. Around the Three Gorge area, the Liantuo Formation attains a maximum thickness of ca. 200 m, unconformably overlying the Huangling granite and disconformably underlying the Nantuo Formation. Chemical Index of Alteration (CIA) of the Liantuo Formation suggests a transition from greenhouse climate conditions during the deposition of its middle-lower part to icehouse conditions during its upper part, foreshadowing the coming of Jiangkou glaciation (Feng et al., 2006; Wang et al., 2006a).

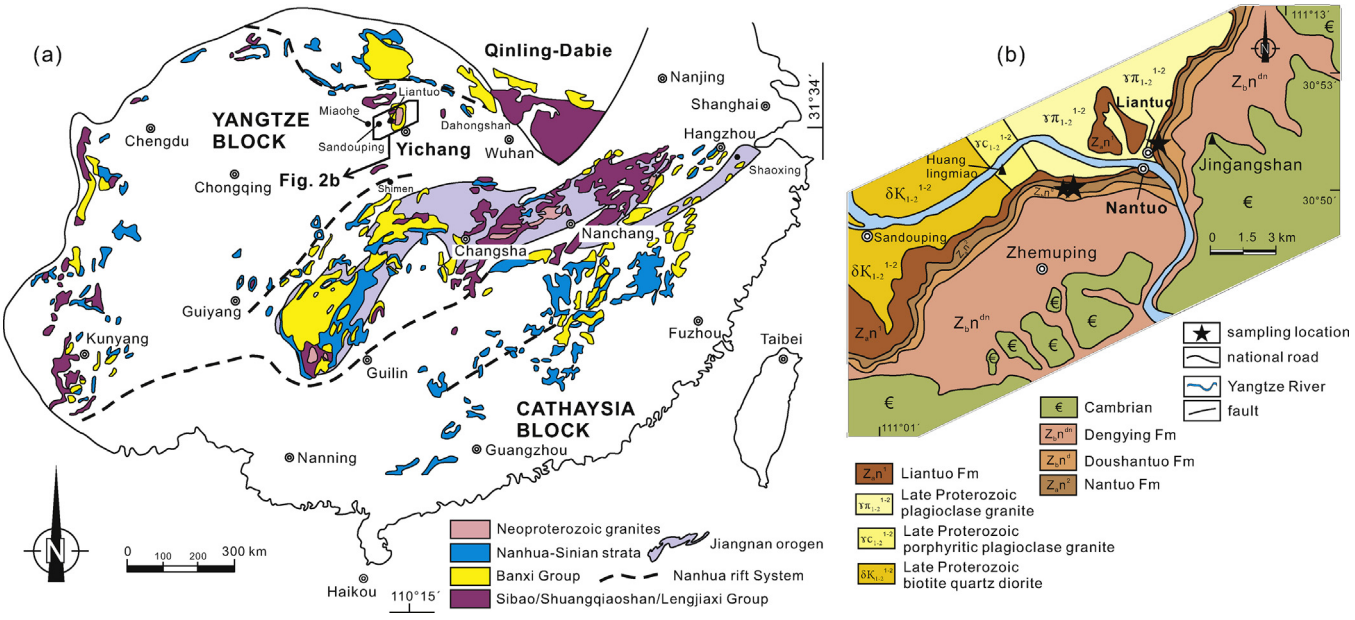


Fig. 1. (a) Outcrops of Neoproterozoic igneous and sedimentary rocks and rift basins in the South China Block (modified from Du et al., 2013). (b) Simplified geological map showing the outcrops of the Liantuo Formation around the study area and sampling locations.

2.2. Previous radiometric dates

The depositional age of the basal Liantuo Formation was previously constrained to be <820 Ma on the basis of SHRIMP U–Pb zircon ages from the Huangling granite around the Three Gorges area (Ma et al., 1984, 7 zircon analyses; Gao and Zhang, 2009, 20 zircon analyses). Recently, Du et al. (2013) obtained a SHRIMP U–Pb zircon age of 779 ± 12 Ma (20 zircon analyses) from a reworked tuff sample within the lower Liantuo Formation in the Dahongshan area, Hubei Province where all the zircon grains have been eroded to varying extent. Ma et al. (1984) reported a SHRIMP U–Pb zircon age of 748 ± 12 Ma (23 zircon analyses) from a tuff sample within the upper Liantuo Formation around the Three Gorges area. Also from the same location, Gao and Zhang (2009) reported a SHRIMP U–Pb zircon age of 724 ± 12 Ma from the upper Liantuo Formation (2 zircon analyses). These radiometric dates suggest depositional age of the top Liantuo Formation should postdate ca. 725 Ma. The Nantuo Formation gave a SHRIMP U–Pb zircon age of ca. 654–635 Ma (Condon et al., 2005, 9 zircon analyses; Zhang et al., 2008c, 19 zircon analyses). Around the Songtao County, Guizhou Province, the basal Datangpo Formation gave a TIMS U–Pb zircon age of ca. 663 Ma (Zhou et al., 2004, 6 zircon analyses) and a SHRIMP U–Pb zircon age of ca. 667 Ma (Yin et al., 2006, 24 zircon analyses). Considering analytical errors and difference in sampling horizons, the basal Datangpo Formation can be constrained to be ca. 670 Ma. Hence available radiometric dates roughly constrained the depositional age of Liantuo Formation to a time interval of 820–670 Ma.

2.3. Sampling details

Representative tuff/tuffaceous siltstone samples were collected from the lower, middle and upper Liantuo Formation from three representative sections at the Wangfenggang, Tianjiayuanzi, and Shibanxi-Gaojialaowu villages, Three Gorges area, Yichang City, Hubei Province (Figs. 1b, and 3). Samples YC01 and YC10, with a distance of about 12 m, were collected from the middle to lower Liantuo Formation at the Wangfenggang section. Sample YC01 is tuffaceous siltstone that has a finely sandy texture, massive structure, and is matrix cemented (Figs. 4a and 5a). It is composed of sandy detritus with subsidiary cements. The sandy detritus is mainly composed of angular to subangular to subrounded feldspar (10–15%, dominated by plagioclase with minor potassic feldspar), quartz (20–25%), and lithic fragments (15–20%, mudstone and biotite) in a dominant size range of 0.05–0.25 mm and a minor size range of 0.3–0.5 mm, with the former and later accounting for 55–60% and 4–5%, respectively. The interstitial materials are mainly muddy materials with a size of <0.05 mm, accounting for 35–40% of the total. Interstitial materials <0.05 mm accounts for 4–5%. Clay matrix accounts for 30–35%. The muddy detritus appear rounded to subrounded morphology with some grains approaching 0.5 mm. Accessory minerals include zircon and pyrite. Some quartz and feldspar appear an angular morphology. Some of the feldspar grains have a clean surface, polysynthetic twin, while others have been strongly sericitized with some residual feldspar. The angular quartz and polysynthetic twin feldspar indicate a volcanic origin.

Sample YC10 is a vitroclastic crystal fragment tuff that has a vitroclastic crystal fragment tuffaceous texture, massive structure (Figs. 4b and 5b). It is composed of crystal fragments, vitroclastics, lithic fragments, and volcanic dust. The crystal fragments are dominated by angular to subangular quartz, feldspar, and biotite in the size range of 0.02–0.2 mm, accounting for 25–30%. The vitroclastics are cryptocrystalline because of devitrification, appearing a needle, arc or chicken bone morphology and accounting for 15–20%

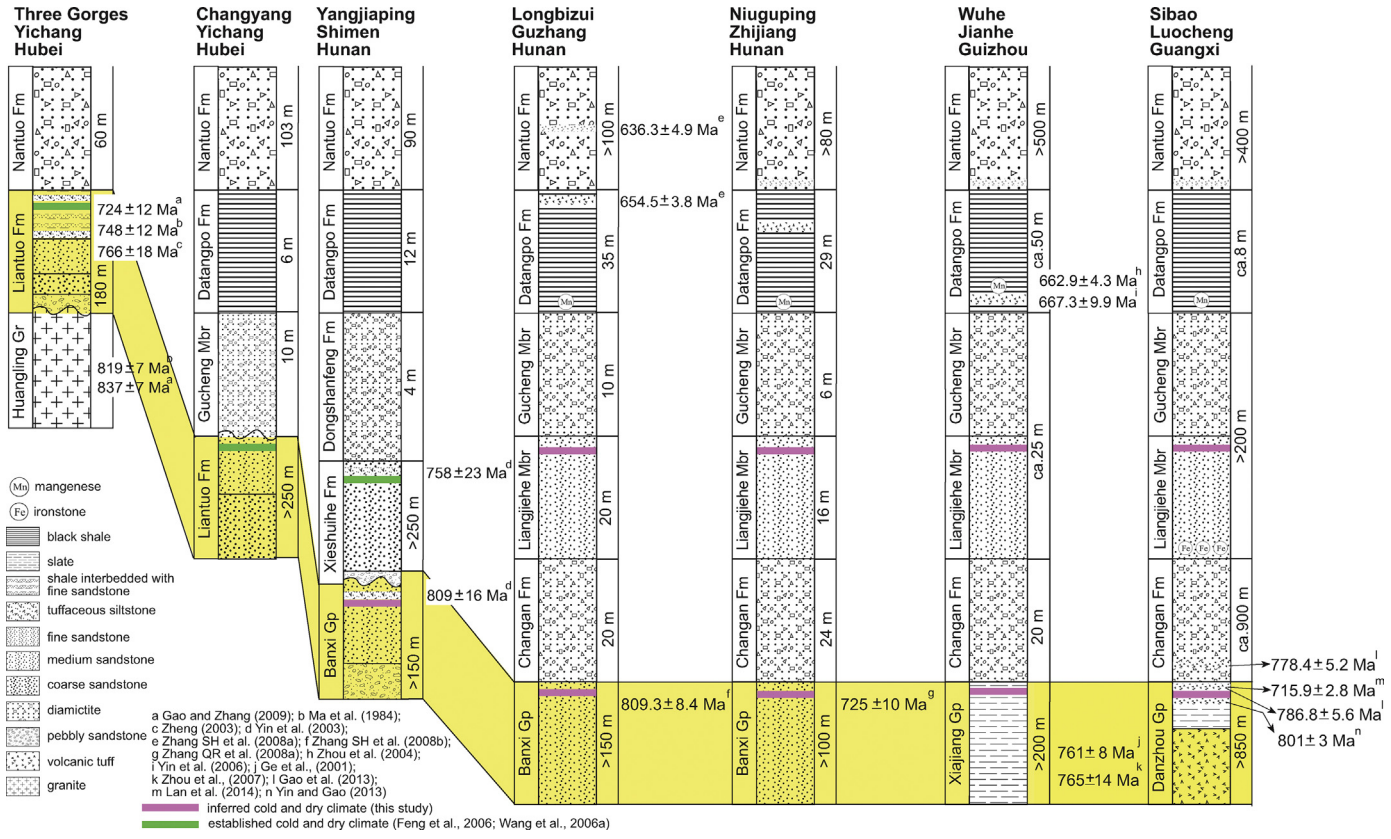


Fig. 2. Schematic stratigraphic columns of the Nanhuan system in the Hubei–Hunan–Guizhou–Guangxi provinces. Also shown are some radiometric dates from tuff/tuffaceous siltstone beds and volcanic rocks.

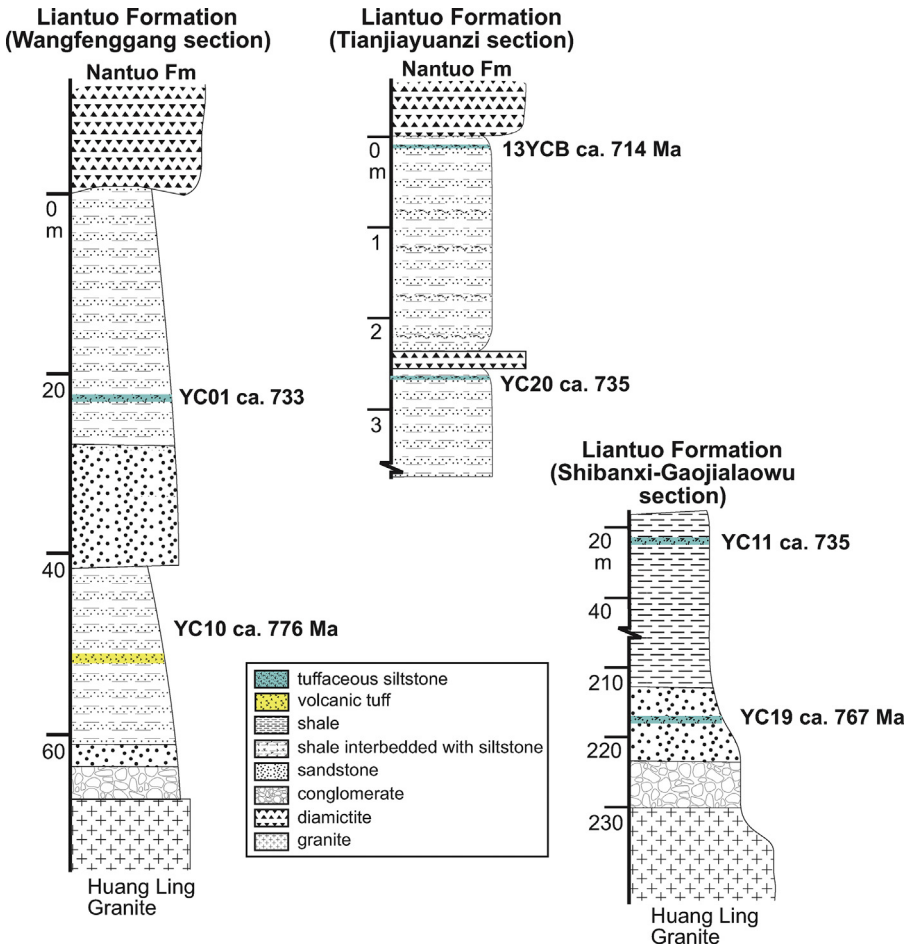


Fig. 3. Stratigraphic columns of the Liantuo Formation at the Tianjiayuanzi, Wangfenggang and Shibanxi-Gaojialaowu sections showing the sampling horizons.

of the total. The lithic fragments are tuffaceous, andesine, basaltic or felsic in the size range of 0.02–0.5 mm, accounting for 5–6%. All the detritus are cemented by volcanic dust. During later diagenesis, devitrification of vitroclastics and volcanic dust occurred. Mafic minerals are partly weathered into muddy materials with iron minerals limonitized.

Samples YC11 and YC19 were collected from the upper and middle Liantuo Formation, respectively at the Shibanxi-Gaojialaowu section. Sample YC11 is a tuffaceous siltstone sample that has a finely sandy texture, massive structure, and is matrix cemented (Figs. 4c and 5c). It is composed of sandy and muddy detritus. The sandy detritus is composed of angular to sub-angular quartz, feldspar and mica, accounting for 10–15% of the total. The muddy detritus is mainly composed of fibrous, cryptocrystalline argillaceous minerals such as kaolinite or montmorillonite with a size <0.005 mm, accounting for 85–90% of the total. Minor iron minerals are sporadically present.

Sample YC19 is a vitroclastic crystal fragment tuff that has a vitroclastic crystal fragment tuffaceous texture, massive structure (Figs. 4d and 5d). It is composed of crystal fragments, vitroclastics, lithic fragments, and volcanic dust. The crystal fragments are dominated by angular to subangular quartz, feldspar, and biotite in the size range of 0.02–0.2 mm, accounting for 15–20%. The vitroclastics are brown cryptocrystalline because of devitrification, appearing a needle, arc or chicken bone morphology and accounting for 10–15% of the total. The lithic fragments are tuffaceous or felsic in the size range of 0.02–0.5 mm, accounting for 4–5%. All the detritus are cemented by volcanic dust. During later diagenesis, devitrification of vitroclastics and volcanic dust occurred.

Sample YC20 and 13YCB are collected from the upper Liantuo Formation at the Tianjiayuanzi section. Of these, sample YC20 was sampled at a distance of 40 cm from the Liantuo-Nantuo Formation boundary, whereas sample 13YCB was sampled 20 cm away from the boundary, the same horizon as that documented by Gao and Zhang (2009). Sample YC20 is silty mudstone or crystal fragment tuff that is dominated by muddy or tuffaceous materials, accounting for 90–95% (Figs. 4e and 5e). Other rock components include silty detritus or crystal fragments that are composed of angular–subangular quartz, feldspar, and mica, 0.01–0.05 mm, accounting for 5–10% of the total.

Sample 13YCB is vitroclastic crystal fragment volcanic tuff composed of vitroclastics, crystal fragments, and volcanic dust (Figs. 4e and 5f). The vitroclastics appear an arc, chicken bone or needle morphology, accounting for 15–20%. The crystal fragments are composed of angular to subangular quartz, plagioclase, and biotite in the size range of 0.02–0.1 mm, accounting for 5–6% of the total. All the detritus are cemented by volcanic dust which account for 75–80%. Some opaque minerals are occasionally present. The mineral compositions define sample 13YCB as tuffaceous siltstone, but not tuff as has been termed by Gao and Zhang (2009).

3. Methods

3.1. SIMS U–Pb analysis

Zircon concentrates were separated from ca. 1.5-kg rock samples by conventional density and magnetic separation techniques. Recovered zircon grains together with chips of zircon standards

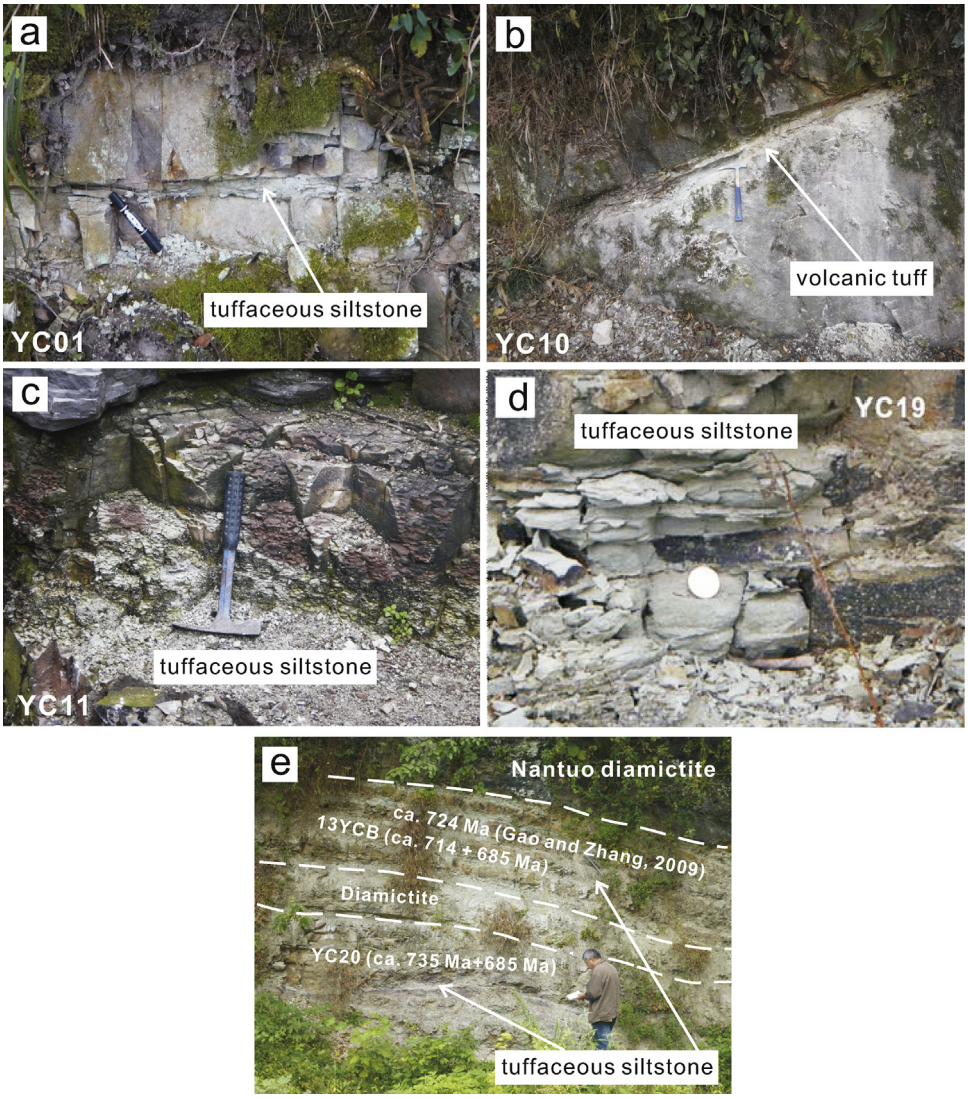


Fig. 4. Field photos showing the sampling sites of tuff/tuffaceous siltstone samples YC01 (a), YC10 (b), YC11 (c), YC19 (d), 13YCB (e), and YC20 (e).

Plešovice and Qinghu were mounted in epoxy discs and polished to expose the longitudinal section of crystals for analysis. All zircons were documented with transmitted and reflected light micrographs as well as cathodoluminescence (CL) and backscattered electron (BSE) images to reveal their internal structures. The SEM instrument is operated with an accelerating voltage of 15 kV. High-quality CL images are acquired at a resolution of 1950 lines/scan at 50 ms duration. A total of 8–12 images are obtained for each sample at magnifications in the range of 130–160× to provide complete coverage of the mounted crystals.

Samples were vacuum-coated with high purity gold and U and Pb isotopes were measured utilizing the Cameca IMS 1280 at the Institute of Geology and Geophysics, Chinese Academy of Sciences (IGGCAS) in Beijing. The SIMS methodology adopted at the IGGCAS follows previously published analytical descriptions with minor modifications (Li et al., 2009b). Each spot was cleaned by rastering the primary ion beam over the target area for about 2 min prior to analysis. Monocollector mode was adopted to collect isotopic data. For normal zircon analyses, the O_2^- primary beam was accelerated at 13 kV with an intensity of about 12 nA. The aperture illumination mode (Kohler illumination) was applied in combination with a 200 μm primary beam mass filter (PBMF) aperture to produce even sputtering over the entire analyzed area. The resulted ellipsoidal spot is about 20 μm × 30 μm in size. For zircon overgrowth analyses, the O_2^- primary beam was accelerated at 13 kV with an intensity of about 1.5 nA. The Kohler illumination was applied combined with a 100 μm PBMF aperture to produce even sputtering over the entire analyzed area. The resulted ellipsoidal spot is about 10 μm × 15 μm in size. Positive secondary ions were extracted with a 10 kV potential. In the secondary ion beam optics, a 60 eV energy window was used together with a mass resolution of about 5400 to resolve lead isotopes from most potential isobaric interferences. The field aperture was set to 5000 μm, and the transfer optic magnification was adjusted to 200. Each measurement consists of 7 cycles, thus the total analytical time is about 12 min.

U–Th–Pb ratios and their absolute abundances were determined in relation to the standard zircon Plešovice (Sláma et al., 2008) and 91500 (Wiedenbeck et al., 1995), respectively. Pb/U calibration was performed in relation to standard zircon Plešovice. Measured compositions were corrected for common lead using non-radiogenic ^{204}Pb. An average present-day crustal composition (Stacey and Kramers, 1975) is used for the common Pb assuming that the common Pb is largely surface contamination introduced during sample preparation. The external uncertainties of SIMS U–Pb measurements were monitored by inter-calibration of standard zircon

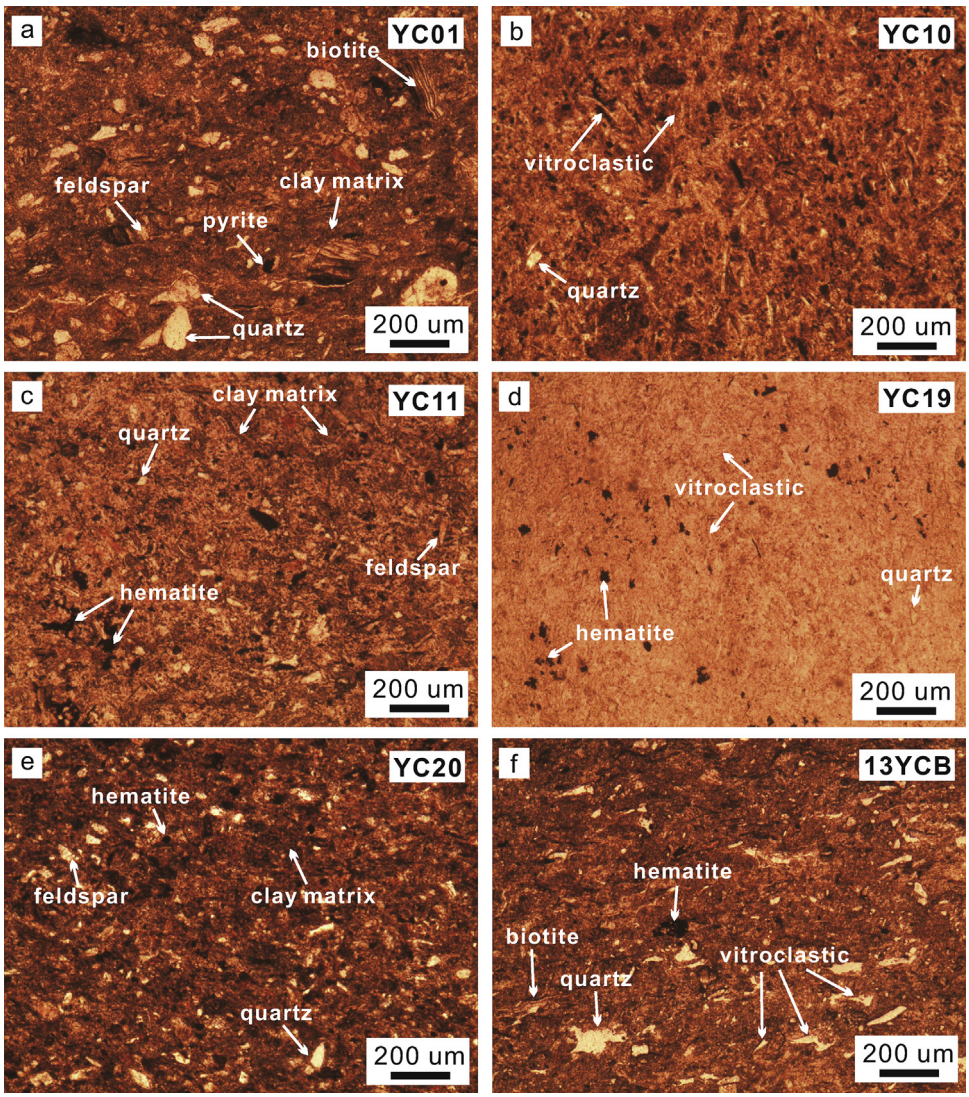


Fig. 5. Photomicrographs of tuff/tuffaceous siltstone samples from the Liantuo Formation. Montage images of reflected and transmitted light.

Plešovice. More details for calibration methods refer to Li et al. (2009b) and Li et al. (2010a). Raw SIMS U–Pb zircon data are reduced using the Cameca IMS 1280 HR built-in software CIPS, and were plotted using the Isoplot 3.7 programs of Ludwig (2008) with decay constants recommended by Steiger and Jäger (1977). Uncertainties on each spot analysis are documented at 1σ level in data tables. For interpretation of the zircon ages only concordant to nearly concordant ($\leq$10% discordant) data were considered unless otherwise specified.

As most of the studied samples are tuffaceous siltstones, a certain amount of detrital zircon grains must have been mixed. This can be shown in the age probability density diagrams with multiple different age groups. In these circumstances, weighted mean dates are determined from each group of analyses by weighting each $^{238}U/^{206}Pb^*$ date by the inverse of its variance (the square of its 1σ analytical uncertainty), and the youngest group of zircon is generally chosen as the best estimate for the volcanic tuff crystallization age, although it is not always this case due to inheritance of zircon grains captured during magma migration (e.g. McCormack et al., 2009). For zircons with normal uranium amount and low common Pb content (e.g. zircons in YC01, YC10, YC11 and 13YCB), the ^{204}Pb-correction method was preferentially adopted to acquire $^{238}U/^{206}Pb^*$ ages, and the conventional U–Pb Concordia diagram (Wetherill, 1956) was used to plot each/single group of zircon ages.

For zircons with low uranium amount and/or high common Pb content (e.g. zircons in YC19 and YC20), the ^{204}Pb-correction method would be inaccurate. Instead, the ^{207}Pb-correction method of common Pb is adopted. This method requires the choice of a regression from initial Pb and assumes that the total Pb is a two-component mixture of common and radiogenic Pb, with both the U–Pb and Pb–Pb isotopic systems being concordant. The raw data will be projected on a mixing line between common $^{207}Pb/^{206}Pb$ and the radiogenic $^{238}U/^{206}Pb^*$ on the inverse Tera-Wasserburg Concordia diagram (Tera and Wasserburg, 1972). The mean $^{238}U/^{206}Pb^*$ age can be calculated based on the ^{207}Pb-correction method of common Pb, which should be consistent within errors with the lower intercept age on the inverse Concordia curve.

3.2. Small beam technique

Some of zircon grains bear bright domains of 5–15 μm in size which is much smaller than the conventional lateral resolution of 20–30 μm can be achieved in the Köhler mode. In order to determine the ages of these bright domains, the Gaussian mode in Cameca IMS 1280 HR instrument is applied for tuning the primary ion beam. The Gaussian illumination mode produces a focused beam density (Liu et al., 2011). As such, a smaller primary beam

with higher ion density can be achieved. The smaller primary beam allows to apply higher magnification transfer settings to ensure higher transmission (~21 cps/ppm/nA) using O_2^- and higher precision (Liu et al., 2011). A 10 kV potential was utilized to extract positive secondary ions. In this study, a primary O_2^- beam in Gaussian mode was used to sputter zircon bright domains generating a flat "U" shape bottom. The primary ion beam was accelerated at −13 kV and finely tuned to less than 0.5 nA.

3.3. SIMS oxygen analysis

Zircon oxygen isotopes were measured using the same Cameca IMS 1280 ion microprobe at IGGCAS. The Cs^+ primary ion beam was accelerated at 10 kV, with an intensity of ca. 2 nA. The Gaussian mode was applied with a primary beam aperture of 200 μm to reduce aberrations. The analytical spot size is ca. 10 μm in diameter. In order to compensate for sample charging during analysis, a normal incidence electron flood gun was applied with homogeneous electron density over a 100 μm elliptical area. A 10 kV potential was used to extract negative secondary ions. A multicollection mode was adopted to measure oxygen isotope compositions. A mass resolution of ca. 2500 can be obtained to measure oxygen isotopes. Each analysis is composed of 20 cycles, with an internal precision better than 0.2‰ (1σ). Detailed analytical procedures and calibration of instrumental mass fractionation refer to Li et al. (2010b, 2013a).

3.4. SIMS REE and trace element analysis

Measurement of REE and trace element was carried out using the Cameca IMS 1280 HR at IGGCAS. Following SIMS oxygen analysis, the mount was cleaned, polished and gold coated prior to SIMS REE and trace element analysis. O^- was used as primary beam to bombarded the sample with a 23 kV energy to stimulate the secondary beam. The target spot was pre-sputtered for about 120 s by an intensity of ~10 nA primary beam to remove surface gold and possible contaminations. The primary beam was then reduced to ~200 pA with a spot size of ~5 μm during analysis. By cycling the magnet field in monocollector mode, the following secondary ions are collected: $^{28}Si^{16}O$, ^{45}Sc, $^{31}P^{16}O$, ^{49}Ti, ^{51}V, ^{55}Mn, $^{40}Ca_2$, ^{89}Y, ^{90}Zr, ^{93}Nb, ^{138}Ba, ^{139}La, ^{140}Ce, ^{141}Pr, ^{145}Nd, ^{147}Sm, ^{153}Eu, ^{158}Gd, ^{159}Tb, ^{163}Dy, ^{165}Ho, ^{167}Er, ^{169}Tm, ^{171}Yb, ^{173}Yb, ^{175}Lu, ^{180}Hf ^{16}O, ^{208}Pb, $^{232}Th^{16}O$, and $^{238}U^{16}O$. Their counting times are 1.04 s, 2.96 s, 2.96 s, 2.96 s, 2.96 s, 2.96 s, 2.96 s, 2.96 s, 2.96 s, 2.96 s, 2.96 s, 8.00 s, 8.00 s, 8.00 s, 10.00 s, 10.00 s, 2.00 s, 2.00 s, 2.00 s, 2.00 s, 2.00 s, 2.00 s, 2.96 s, 2.96 s, 2.96 s, 2.96 s, 2.96 s, 2.96 s, 2.96 s, and 2.96 s, respectively. Each spot analysis was composed of five cycles. The signals were collected on electron multiplier. The $^{28}Si^{16}O$ ions were used to center the secondary beam and monitor instrumental drift. An entrance slit width of 40 μm, a field aperture 3000 μm and an energy slit width of 50 eV were applied. With the above configuration, a mass resolution of ~13,000 (50% peak height definition) can be achieved to separate interference. A transfer magnification of ~300 was used to improve the transmission of secondary beam. M257 zircon standard (Nasdala et al., 2008) and NIST610 (Pearce et al., 1997) were used to correct REE and trace element compositions of the raw data, respectively.

4. Results

4.1. Zircon SIMS U–Pb ages

SIMS U–Pb analytical results of 272 zircon grain analyses from samples YC01, YC10, YC11, YC19, YC20, and 13YCB are listed in Supplement 1. One sigma error was quoted for individual ratios and ages. $^{206}Pb/^{238}U$ ages are more precise and reliable than $^{207}Pb/^{206}Pb$ ages for zircons younger than 1000 Ma (Li et al., 2009b; Lan and Chen, 2012). All ages are <1000 Ma in this study, and thus $^{206}Pb/^{238}U$ ages were used to calculate the population age of all zircons unless otherwise specified. Zircon grains from sample YC01 are transparent, colorless in color, 50–230 μm long, 30–80 μm wide with a length/width (l/w) ratio ranging from 1.3 to 3.8 (Fig. 6). The apparent morphology ranges from euhedral and long-prismatic to stubby, subangular grains. Of these, the euhedral grains are the most common. CL images reveal that most of the zircons, especially those bright color ones, show distinct oscillatory zoning characteristics, and are interpreted as magmatic in origin (Zeck and Williams, 2002; Corfu et al., 2003). In SIMS U–Pb analysis, only those prismatic, euhedral zircons with oscillatory zoning were selected. A total of 32 ages were obtained from sample YC01 which extend from 802 Ma to 731 Ma with a major peak at ca. 774 Ma (Fig. 7a). The ca. 774 Ma peak is produced by thirty zircon grain analyses which yielded a weighted mean $^{206}Pb/^{238}U$ age of 771.9 ± 4.6 Ma (MSWD = 2.4) and a Concordia age 771.7 ± 3.7 Ma (MSWD = 0.04) (Fig. 7b). The Concordia age means no or less radiogenic Pb loss once their U–Pb isotope systems were closed. In this case, the $^{206}Pb/^{238}U$ age approximates $^{207}Pb/^{235}U$ and $^{207}Pb/^{206}Pb$ ages. Zircon grains yielding the ca. 771 Ma age commonly display subhedral morphology with visible oscillatory or sector zoning (e.g. Fig. 6, YC01@17, YC01@19, YC01@20). We interpreted the ca. 771 Ma age has derived from sedimentary recycled zircon grains in reworked volcanic tuff. Two zircon grains yielded a weighted mean $^{206}Pb/^{238}U$ age of 732 ± 10 Ma (MSWD = 0.06). Zircon grains producing this age show typical euhedral morphology with a distinct oscillatory zoning (e.g. Fig. 6, YC01@01, YC01@02). Hence we interpreted ca. 733 Ma as the most approximate depositional age of sampling horizon.

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Zircon grains from samples YC10 are transparent, colorless to light pink in color, 45–185 μm long, 40–110 μm wide with a length/width (l/w) ratio ranging from 1.1 to 3.2 (Fig. 6). The apparent morphology ranges from euhedral long-prismatic to semi-prismatic grains, but is dominated by the former. CL images show that almost all the zircons exhibit distinct oscillatory zoning characteristics indicative of magmatic origin (Zeck and Williams, 2002; Corfu et al., 2003). A total of 33 analyses were acquired from sample YC10 which constitute a single group (Fig. 8a) and yielded a weighted mean $^{206}Pb/^{238}U$ age of 776.6 ± 3.8 Ma (MSWD = 2.1) and a Concordia age of 776.5 ± 3.2 Ma (MSWD = 0.01) (Fig. 8b). Zircon grains yielding the ca. 776 Ma age exhibit typical euhedral morphology with distinct oscillatory zoning, typical of magmatic zircons (Fig. 6, YC10). Hence we interpret ca. 776 Ma as the most approximate depositional age of sampling horizon.

Zircon grains from sample YC11 are transparent, colorless to opaque in color, 28–110 μm long, 25–60 μm wide with a length/width (l/w) ratio ranging from 1.1 to 2.0 (Fig. 6). The apparent morphology ranges from prismatic to semi-prismatic to subrounded grains. Of these, the prismatic to semi-prismatic grains are the most common. CL images reveal that most of the prismatic to semi-prismatic grains show distinct oscillatory zoning characteristics suggestive of magmatic origin (Zeck and Williams, 2002; Corfu et al., 2003). A total of 57 analyses were acquired from sample YC11 which extend from 875 Ma to 724 Ma and fall into two major peaks at ca. 764 Ma and ca. 738 Ma (Fig. 9a). The major age peak at ca. 764 Ma is produced by 47 grain analyses which yielded a weighted mean $^{206}Pb/^{238}U$ age of 767.2 ± 3.6 Ma (MSWD = 2.1) and a Concordia age 767.4 ± 2.9 Ma (MSWD = 1.2) (Fig. 9b and c). Zircon grains yielding this age exhibit subangular morphology with visible oscillatory zoning (e.g. Fig. 6, YC11@43, YC11@45), indicating transportation and erosion to some extent. The second age peak at ca. 738 Ma is produced by nine grain analyses which yielded

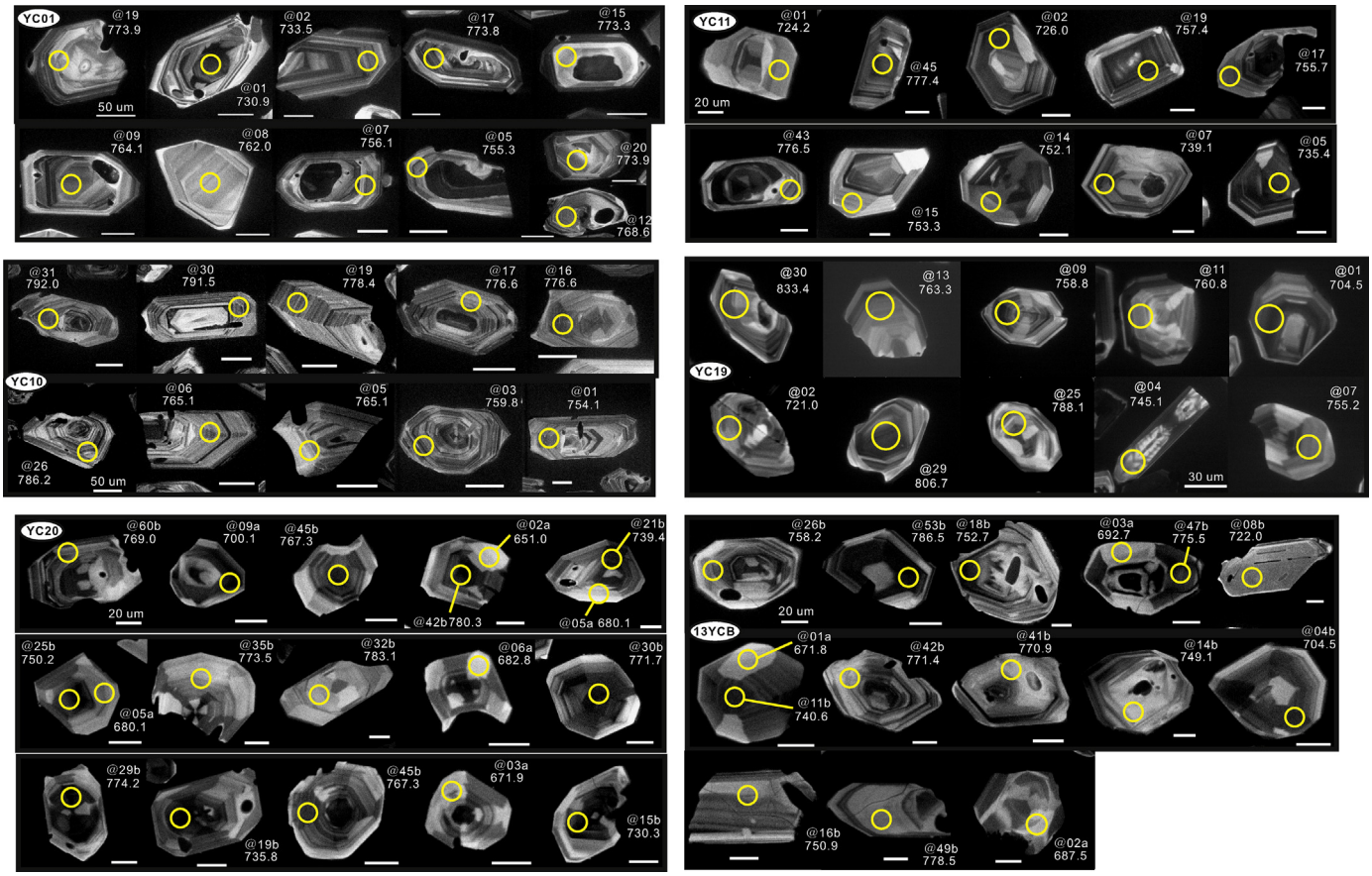


Fig. 6. CL images of zircon from the tuff/tuffaceous siltstone samples from the Liantuo Formation. In sample YC01 and YC10, all the scales represent 50 μm. In samples YC11, YC20 and 13YCB, all the scales represent 20 μm, whereas in sample YC19, all the scales represent 30 μm.

a weighted mean $^{206}Pb/^{238}U$ age of 734.7 ± 5.7 Ma (MSWD = 0.54) and a Concordia age 735.5 ± 5.8 Ma (MSWD = 4.2) (Fig. 9d and e). Zircon grains yielding this age exhibit euhedral morphology with visible oscillatory or sector zoning (e.g. Fig. 6, YC11@05, YC11@07, YC11@14). As such, we interpret the ca.735 Ma as the most approximate depositional age of the sampling horizon.

Zircon grains from sample YC19 are transparent, colorless to light pink in color, 60–80 μm long, 25–55 μm wide with a length/width (*l*/*w*) ratio ranging from 1.1 to 3.2 (Fig. 6). The apparent morphology ranges from prismatic to stubby to subrounded grains. Of these, the euhedral, stubby grains are the most common. CL images reveal that most of the prismatic to stubby grains show oscillatory zoning characteristics indicative of magmatic origin (Zeck and Williams, 2002; Corfu et al., 2003). In contrast, all the subrounded grains show opaque color suggestive of an inherited origin. A total of 29 analyses were obtained from sample YC19 which extend from 832.6 Ma to 740.3 Ma and fall into two major peaks at ca. 792 Ma and ca. 768 Ma (Fig. 10a). The major age peak at ca. 768 Ma is produced by 22 grain analyses. In the Terra-Wasserburg plot, these 22 analyses give a lower intercept age of 767.4 ± 3.4 Ma (MSWD = 1.15) (Fig. 10b). The second age peak at ca. 792 Ma is produced by six grain analyses. Zircon grains yielding the age of ca.

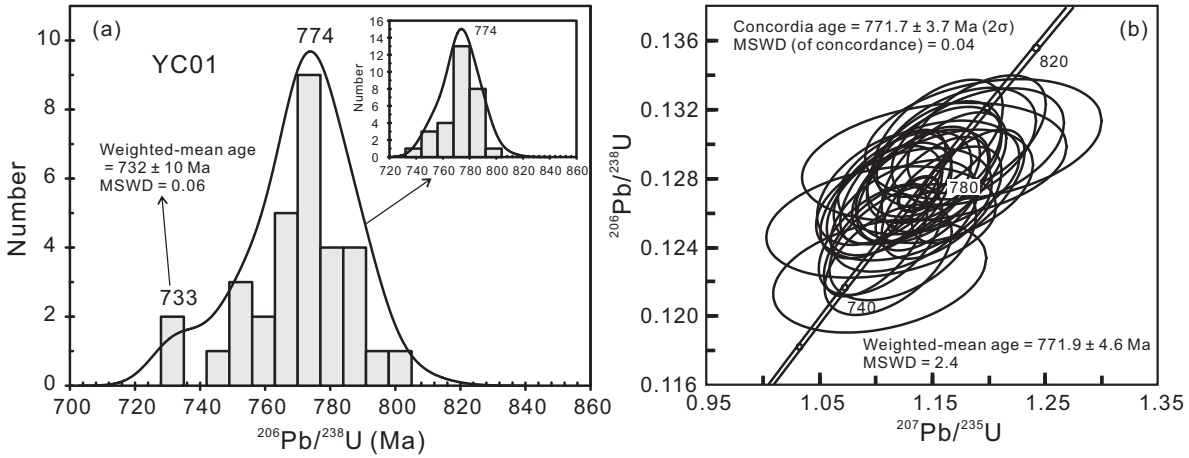


Fig. 7. Sample YC01 age population plot (a) and Concordia plot (b).

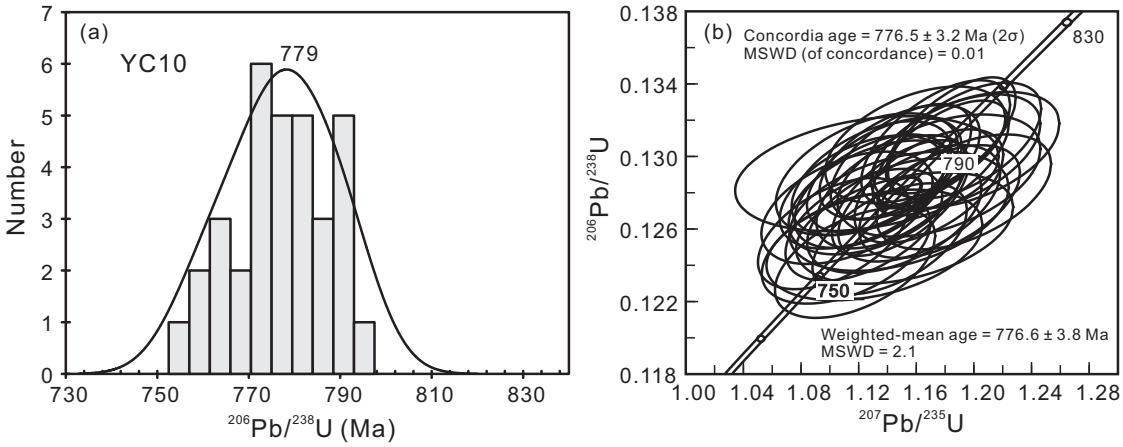


Fig. 8. Sample YC10 age population plot (a) and Concordia plot (b).

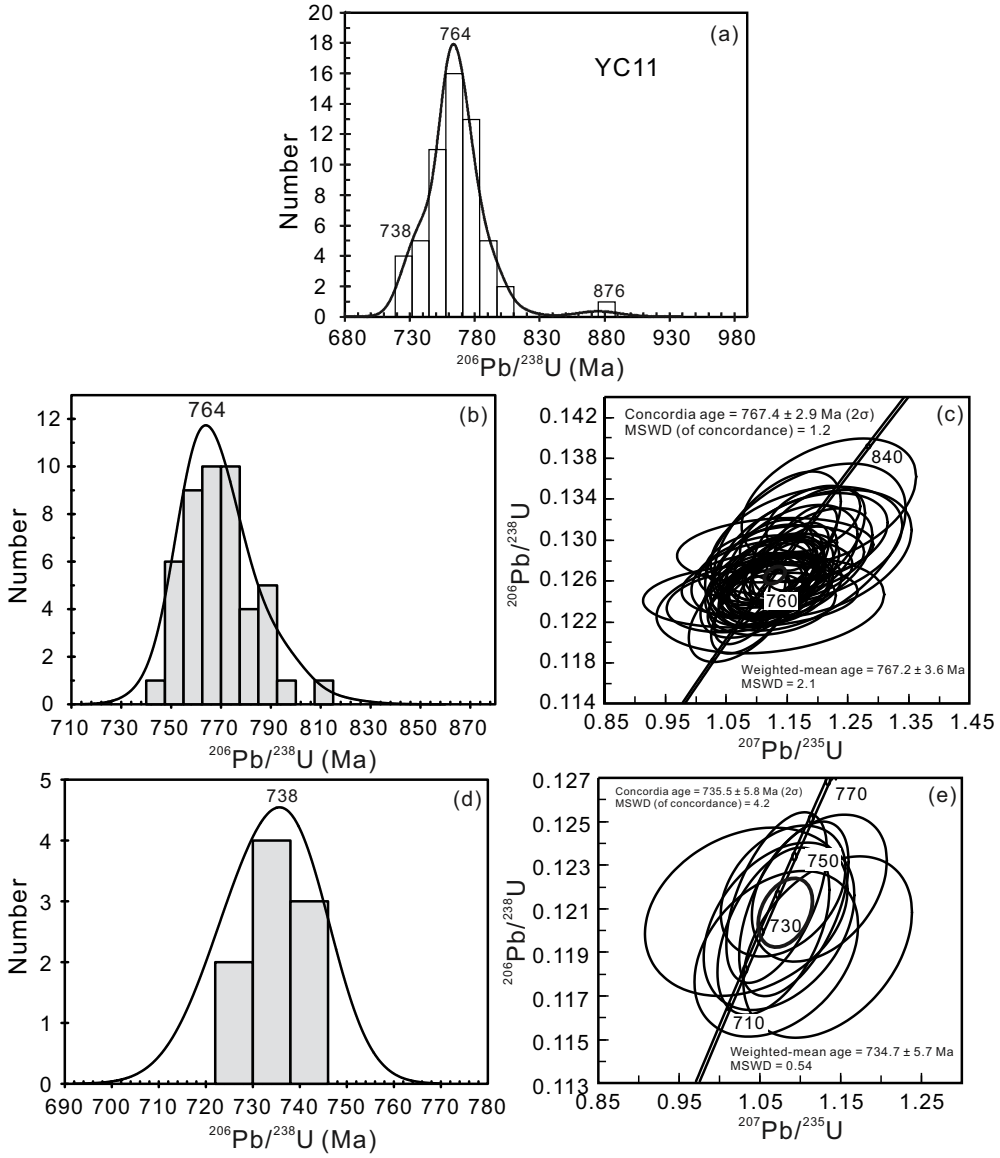


Fig. 9. Sample YC11 age population plots (a, b, d) and Concordia plots (c, e).

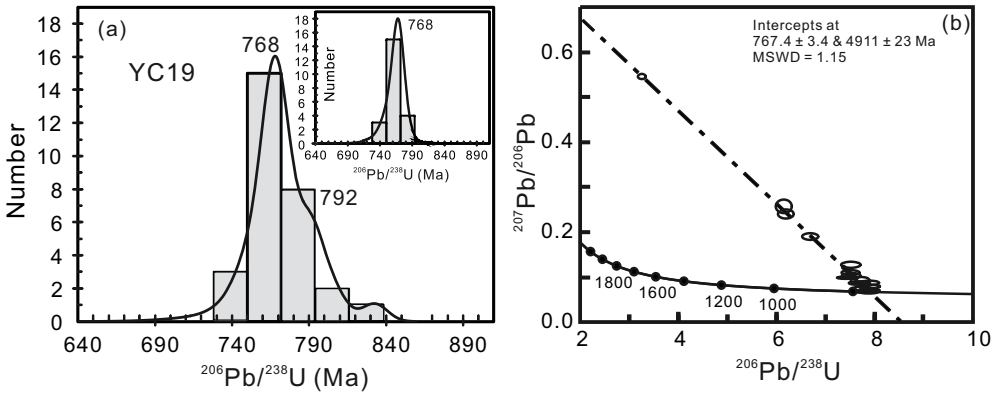


Fig. 10. Sample YC19 age population plot (a) and Tera and Wasserburg plot (b).

767 Ma show typical euhedral morphology with distinct oscillatory zoning (e.g. YC19@01, YC19@02, YC19@04, YC19@07), whereas zircon grains producing the ages of ca. 792 Ma exhibit subangular morphology with indistinct oscillatory zoning (e.g. YC19@25, YC19@29, YC19@30). Hence we interpret the ca. 768 Ma as the most approximate depositional age of the sampling horizon, all the other ages being derived from sedimentary recycled zircon grains in reworked volcanic tuff.

Zircon grains from sample YC20 are transparent, colorless to opaque in color, 25–105 μm long, 25–65 μm wide with a length/width (*l*/*w*) ratio ranging from 1.1 to 3.5 (Fig. 6). The apparent morphology ranges from prismatic to stubby to subrounded grains. CL images reveal that most of the prismatic to stubby grains show oscillatory zoning characteristics suggestive of magmatic origin (Zeck and Williams, 2002; Corfu et al., 2003). In contrast, all the subrounded grains show opaque color, indicating an inherited origin. Some of the prismatic to stubby grains bear bright domains that either surrounded or truncated the cores. A total of 63 analyses were obtained from sample YC20 which extend from 754.5 Ma to 646.7 Ma and fall into three major age groups peaked at ca. 764 Ma, ca. 736 Ma, and ca. 685 Ma (Fig. 11a). The major age peak at ca. 764 Ma is represented by 40 grain analyses which yielded a weighted mean ^{206}Pb/^{238}U age of 762.2 ± 3.4 Ma (MSWD = 2.0) and a Concordia age 762.6 ± 2.5 Ma (MSWD = 5.6) (Fig. 11b and c). The second age peak at ca. 736 Ma is represented by 13 grain analyses and yielded a lower intercept age of 735.2 ± 5.8 Ma (MSWD = 0.31) in the Tera-Wasserburg plot and a weighted mean ^{207}Pb corrected ^{206}Pb/^{238}U age of 738.3 ± 4.9 Ma (MSWD = 0.32) (Fig. 11d and e). The third age peak at ca. 685 Ma is presented by 10 grain analyses and yielded a lower intercept age of 688 ± 15 Ma (MSWD = 1.9) in the Tera-Wasserburg plot and a weighted mean ^{207}Pb corrected ^{206}Pb/^{238}U age of 684.3 ± 9.8 Ma (MSWD = 1.14) (Fig. 11f and g). Zircon grains yielding the ages of ca. 685 Ma and ca. 735 Ma exhibit euhedral morphology with distinct oscillatory zoning (e.g. Fig. 6, YC20@02, YC20@03, YC20@05, YC20@06, YC20@15, YC20@19, YC20@25), whereas zircon grains yielding ca. 762 Ma show subangular morphology (e.g. Fig. 6, YC20@29, YC20@60). All the ages producing the lower intercept age ca. 685 Ma are derived from bright domains of zircon grains, and are highly discordant with high common Pb content, and thus are interpreted as indication of a post-depositional recrystallization event (see discussion below). As such, the ca. 735 Ma is regarded as the best estimation for the depositional age of the sampling horizon.

Zircon grains from sample 13YCB are colorless. Half of them are transparent, and the remainders are translucent to opaque. Size of the zircon grains is 35–135 μm long, 26–100 μm wide with a length/width (*l*/*w*) ratio ranging from 1.1 to 4.8 (Fig. 6). The apparent morphology ranges from long, prismatic to stubby to rounded grains. Of these, the euhedral, stubby grains are the most common. CL images reveal that most of the prismatic to stubby grains show oscillatory zoning characteristics, indicating a magmatic origin (Zeck and Williams, 2002; Corfu et al., 2003). In contrast, the subrounded to rounded grains mostly show opaque color, indicating an inherited origin. Some of the prismatic to stubby grains bear bright domains that either surrounded or truncated the cores. In SIMS U–Pb analysis, both the euhedral zircons and the bright domains were selected. A total of 58 analyses were acquired from sample 13YCB which extend from 799.4 Ma to 671.8 Ma and fall into two major groups peaked at ca. 762 Ma and ca. 714 Ma, respectively (Fig. 12a). The peak at ca. 762 Ma is composed of thirty-three grain analyses which yield a weighted mean ^{206}Pb/^{238}U age of 766.2 ± 4.2 Ma (MSWD = 1.8) and a Concordia age 766.4 ± 3.8 Ma (MSWD = 1.3) (Fig. 12b). The age peak at ca. 714 Ma is composed of seven grain analyses which give a weighted mean ^{206}Pb/^{238}U age of 714 ± 8 Ma (MSWD = 0.81) (Fig. 12c and d). Zircon grains yielding ca. 766 Ma all exhibit a subangular morphology with visible oscillatory or sector zoning (e.g. Fig. 6, 13YCB@08, 13YCB@14, 13YCB@18, 13YCB@26, 13YCB@33, 13YCB@42), and thus are interpreted to have derived from sedimentary recycled zircon grains in reworked volcanic tuff. Instead, zircon grains yielding the age of ca. 714 Ma exhibit euhedral morphology with distinct oscillatory zoning (e.g. Fig. 6, 13YCB@01, 13YCB@02, 13YCB@03, 13YCB@04). As such, we interpret the ca. 714 Ma as the most approximate depositional age of the sampling horizon. The three ages of 671.8 ± 10.4 Ma, 687.5 ± 9.8 Ma, and 692.7 ± 9.9 Ma were derived from the bright domains of zircon grains. Combining these three ages yielded a weighted mean ^{206}Pb/^{238}U age of 684 ± 12 Ma (MSWD = 1.13) and is also interpreted as an indication of post-depositional recrystallization event (see discussion below). To summarize, zircon grains from six tuff/tuffaceous siltstone beds within the Liantuo Formation gave depositional ages of ca. 780–770 Ma, ca. 735–730 Ma, and ca. 714 Ma, respectively, which constrained the depositional age of Liantuo Formation to a time interval of 780–714 Ma. The 690–680 Ma ages could suggest a post-depositional recrystallization event.

4.2. Zircon oxygen isotopes

A total of 32 SIMS oxygen isotope analyses were carried out on samples YC20 to distinguish the genesis of bright domains overgrown on zircon grains. The results are listed in Supplement 2. Among the 32 analyses, twelve were carried out on the bright domains of zircon grains, whereas twenty were conducted on the cores as a comparison. Zircon cores show variable $\delta^{18}O$ values of 1.21‰ to 4.66‰ (Fig. 13a). Zircon bright domains show a similar range of $\delta^{18}O$ values varying from 1.92‰ to 3.91‰ (Fig. 13b).

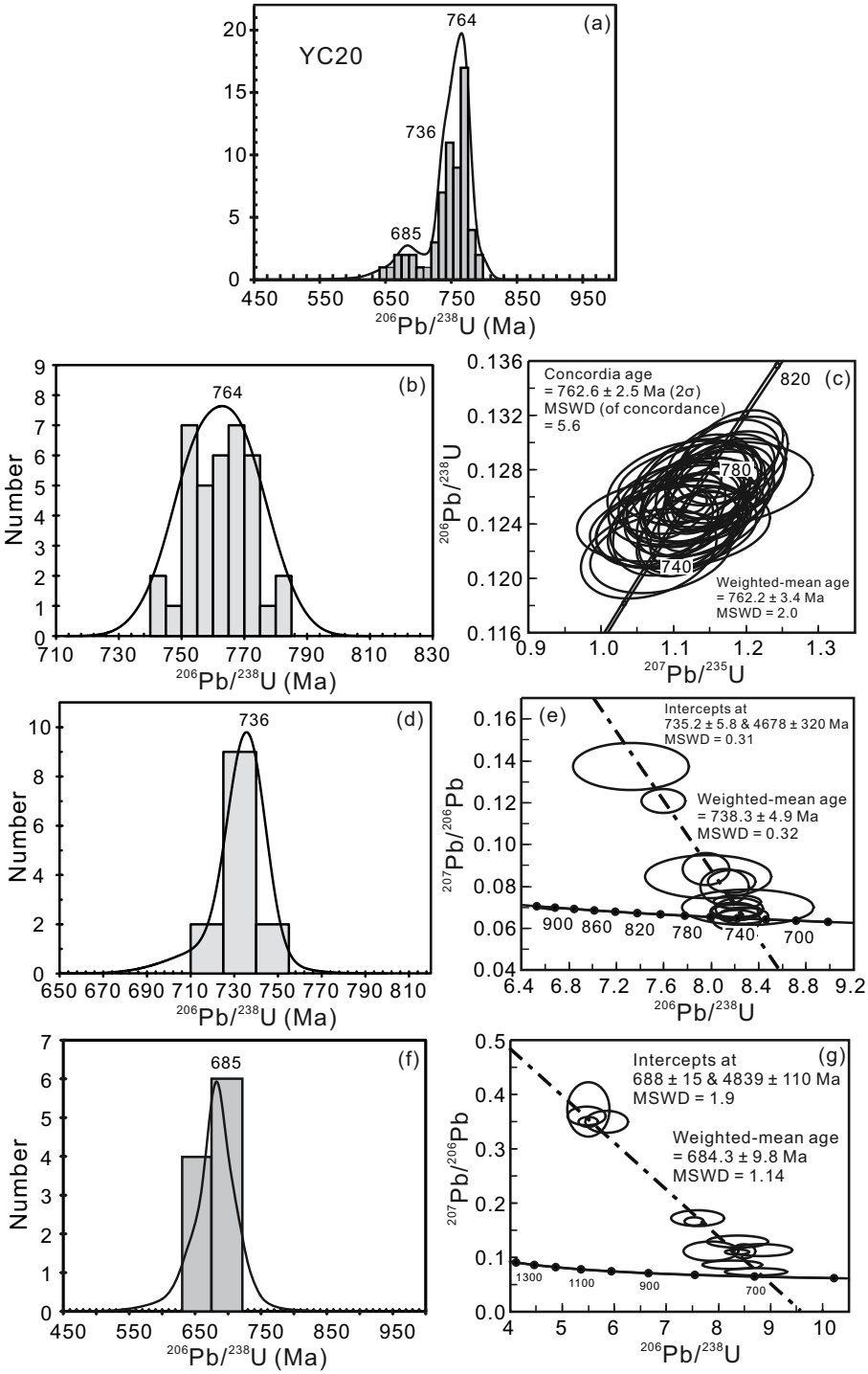


Fig. 11. Sample YC20 age population plot (a, b, d, f), Concordia plot (c), and Tera and Wasserburg plot (e, g).

Obviously, both cores and bright domains are depleted in ^{18}O as their $\delta^{18}O$ values are lower than the normal mantle zircon values of $5.3 \pm 0.3‰$ (Valley et al., 1998).

Supplementary material related to this article can be found, in the online version, at http://dx.doi.org/10.1016/j.precamres.2015.03.012.

4.3. Zircon REE and trace elements

A total of 42 and 66 SIMS REE and trace element analyses were carried out on samples YC20 and 13YCB, respectively, to distinguish the genesis of bright domains of zircon grains. The results are listed in Supplement 3. Both the cores and bright domains of sample YC20 display a left-dipping REE distribution pattern with positive Ce and Eu anomalies (Fig. 14a and c). δCe (Ce/Ce*) and δEu (Eu/Eu*) in bright domains vary from 1.55 to 19.36 with an average of 11.01 and from 0.46 to 2.24 with an average of 1.56, respectively. In contrast, δCe (Ce/Ce*) and δEu (Eu/Eu*) in cores vary from 0.35 to 32.69 with an average of 10.09 and from 0.35 to 1.94 with an average of 1.32, respectively. Likewise, both the cores and bright domains of sample 13YCB display a similar left-dipping REE distribution pattern with positive Ce and Eu anomalies (Fig. 14b and d). δCe (Ce/Ce*)

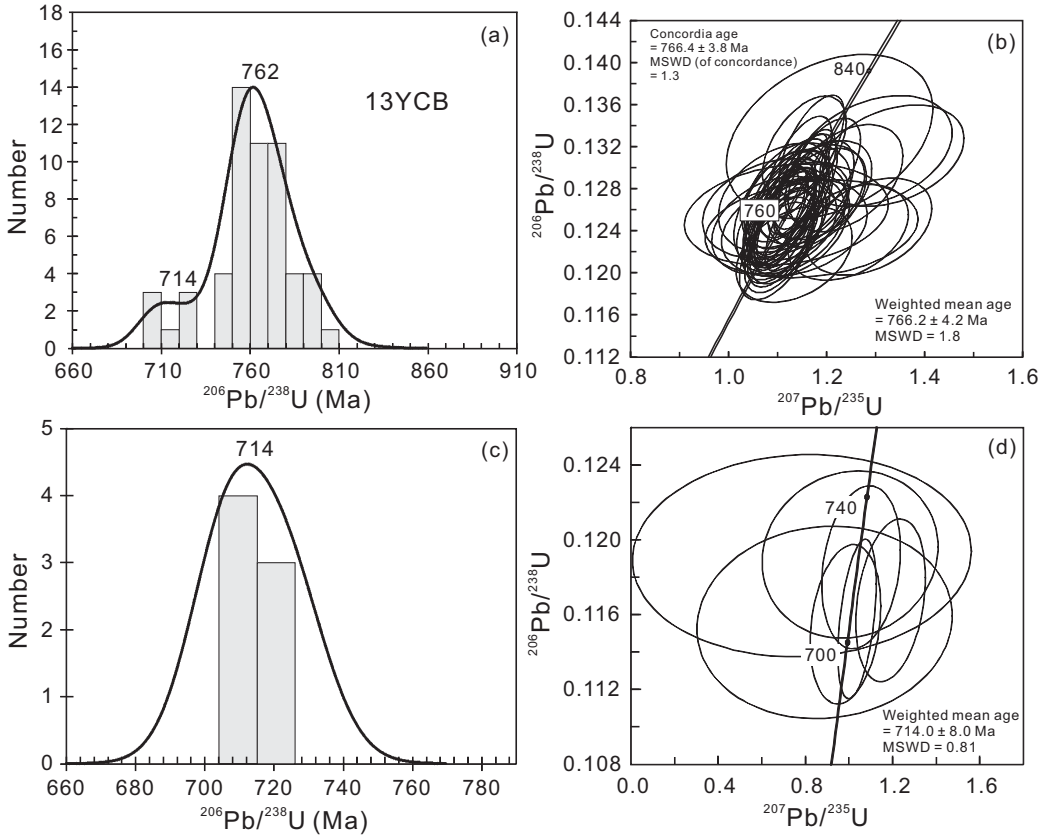


Fig. 12. Sample 13YCB age population plot (a, c) and Concordia plots (b, d).

and δEu (Eu/Eu*) in bright domains vary from 0.82 to 20.82 with an average of 9.68 and from 0.62–2.79 with an average of 1.58, respectively. In contrast, δCe (Ce/Ce*) and δEu (Eu/Eu*) in cores vary from 0.11 to 26.88 with an average of 10.01 and from 0.72 to 2.36 with an average of 1.28, respectively. The zircon cores and bright domains of sample YC20 attains a mean ΣREE of 2635.61 ppm and 859.90 ppm, respectively. In comparison, zircon cores and bright domains of sample 13YCB attains a mean ΣREE of 2707.67 ppm and 626.84 ppm, respectively. As such, both sample YC20 and 13YCB show a lower ΣREE in zircon bright domains relative to zircon cores (Fig. 14e and f).

Supplementary material related to this article can be found, in the online version, at http://dx.doi.org/10.1016/j.precamres.2015.03.012.

Both zircon bright domains and cores of sample YC20 are depleted in Ba, La, Pr, Dy, Ti and enriched in Pb, Ce, Th, Hf (Fig. 15a and c). Similar characteristics are also manifested in zircon bright domains and cores of sample 13YCB (Fig. 15b and d). Both samples YC20 and 13YCB show lower total trace element contents in zircon bright domains relative to zircon cores (Fig. 15e and f). Th/U ratios in the bright domains of sample YC20 vary from 0.58 to 1.77 with an average of 0.93, whereas those in the cores of sample YC20 range from 0.65 to 3.02 with a mean of 1.41 (Fig. 16a). Th/U ratios in the bright domains of sample 13YCB vary from 0.52 to 2.10 with an average of 0.81, whereas those in the cores of sample 13YCB range from 0.46 to 2.96 with a mean of 1.52 (Fig. 16b). It is obvious that both samples YC20 and 13YCB show higher Th/U ratios in zircon cores than bright domains.

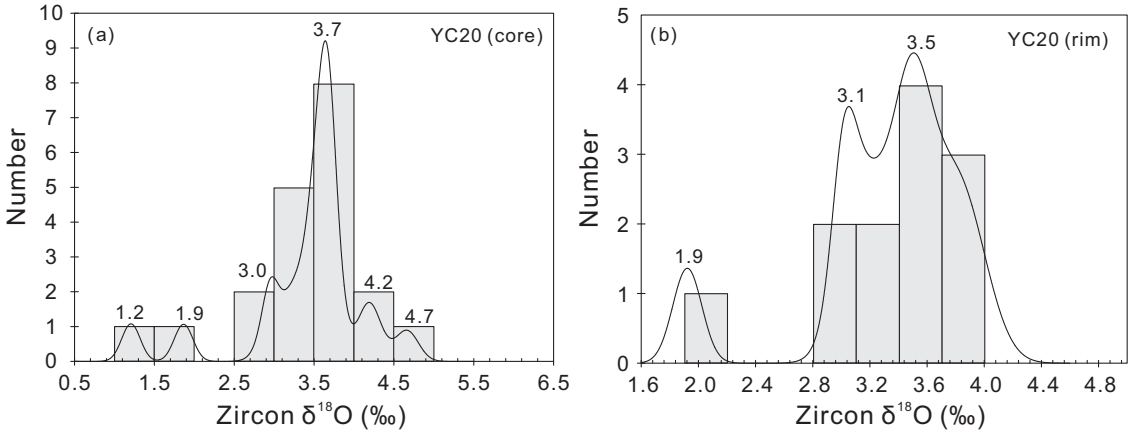


Fig. 13. Oxygen compositions of zircon cores (a) and bright domains (b) in sample YC20.

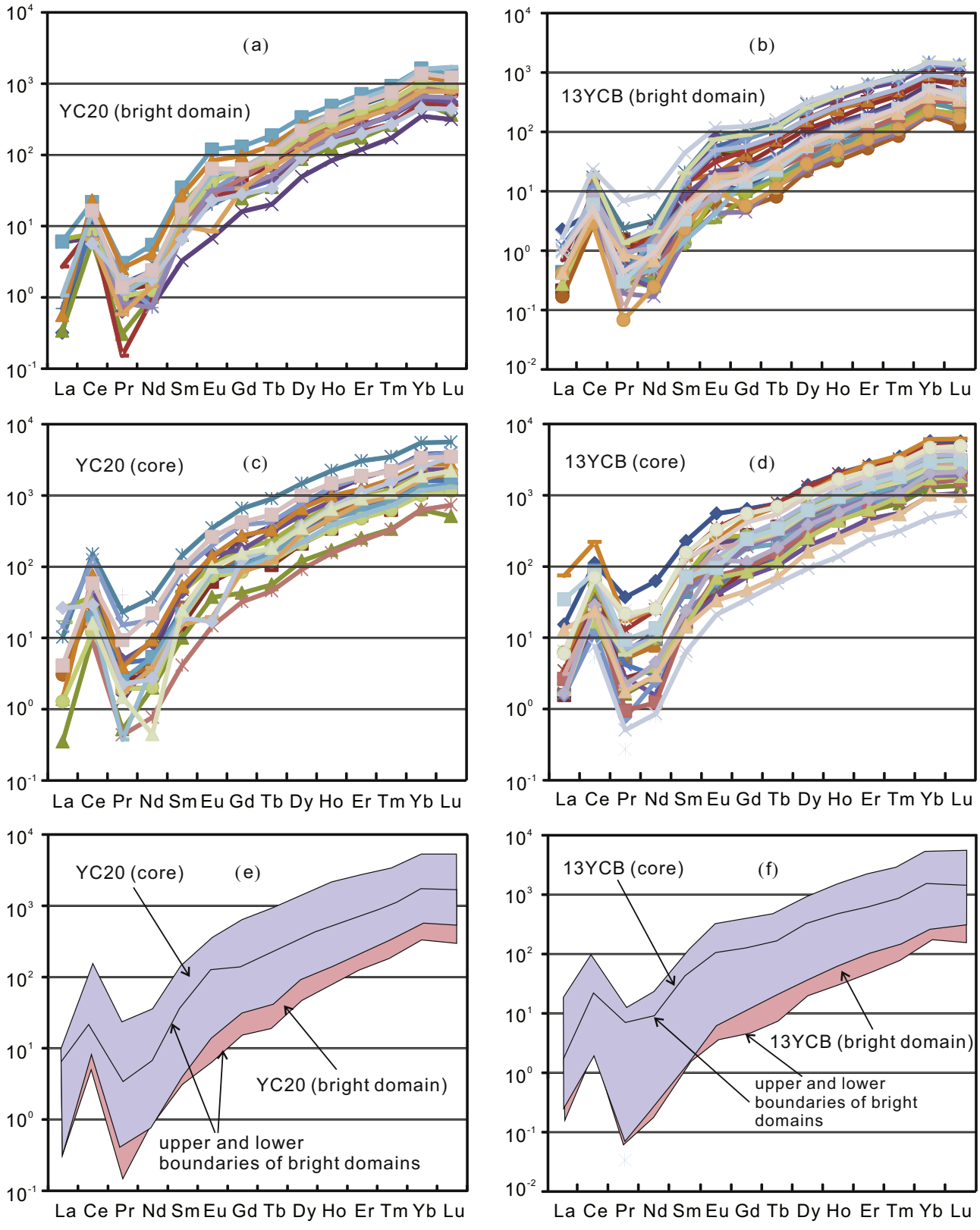


Fig. 14. REE distribution patterns of zircon cores and bright domains and their comparison in samples YC20 (a, c, e) and 13YCB (b, d, f).

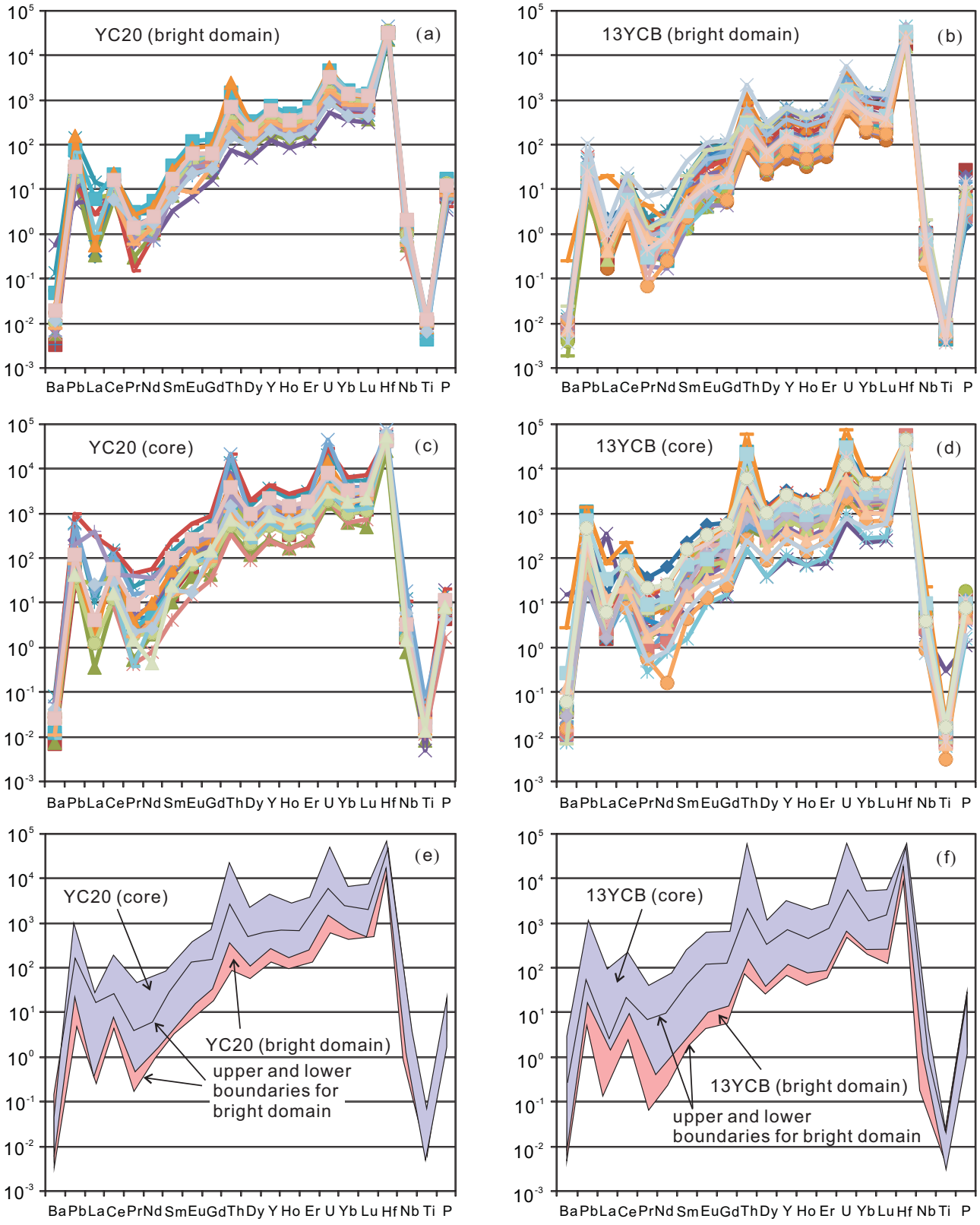


Fig. 15. Trace element distribution patterns of zircon cores and bright domains and their comparison in samples YC20 (a, c, e) and 13YCB (b, d, f).

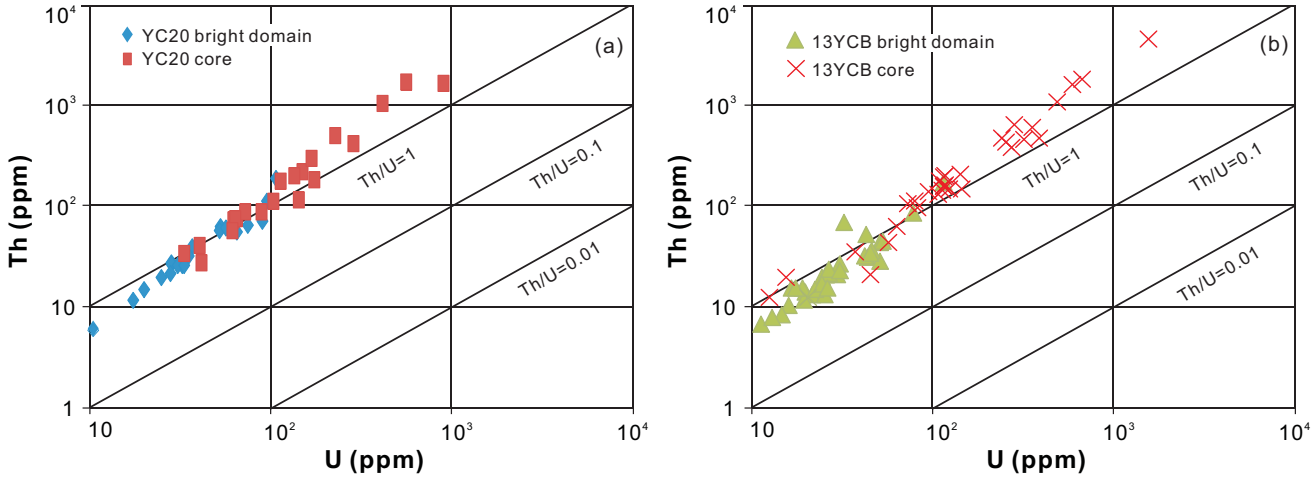


Fig. 16. Th/U ratios of zircon cores and bright domains in samples YC20 (a) and 13YCB (b).

5. Discussions

5.1. Genesis of the bright domains of zircon grains

The bright domains overgrown on zircon grains in sample YC20 and 13YCB from the upper Liantuo Formation yield a group of youngest age of ~680 Ma, which have never been reported in previous studies. Understanding the genesis of these bright domains is crucial to the chronological framework of the Liantuo Formation. Two potential mechanisms could account for the formation of bright zircon domains, i.e. hydrothermal alteration or recrystallization. Hydrothermal zircons are characterized by porous or spongy structures exhibiting a dark CL color, and usually contain abundant inclusions such as xenotime and thorite suggestive of hydrothermal alteration of precursor zircon grains (Hoskin, 2005). Particularly, hydrothermal fluids are rich in LREE and their interaction with zircon grains would result in LREE enrichment in bright domains relative to their igneous cores (Hoskin, 2005). All zircon bright domains of samples YC20 and 13YCB show a bright CL and a smooth surface containing little inclusions. Neither HREE nor LREE contents is higher in zircon bright domains relative to their cores, indicating a hydrothermal alteration process is impossible. Furthermore, fluid involved hydrothermal alteration would change the oxygen isotope compositions of zircon grains, shifting it toward a much lower value (Zheng et al., 2004; Wu et al., 2007). The $\delta^{18}O$ values of zircon bright domains of samples YC20 and 13YCB are indistinguishable from cores, again indicating a hydrothermal alteration process is impossible.

The zircon bright domains of samples YC20 and 13YCB commonly grow nearly perpendicular to the directions of growth bands and sector boundaries and follow the contours of precursor zircon grains suggestive of recrystallization of precursor zircon grains (c.f. Vavra et al., 1996). Recrystallization has made the internal zoning structures of some zircon grains missing or blurred (e.g. Fig. 6, YC20@09, YC20@29, 13YCB@01, 13YCB@02, 13YCB@03, 13YCB@08, 13YCB@47, 13YCB@53) despite preservation of magmatic oscillatory zonation patterns in the cores of some zircon grains, suggesting varying extent of recrystallization modification on magmatic domains (Chen et al., 2010). A recrystallization process is also evidenced by their consistent $\delta^{18}O$ values with the igneous domains, high Th/U ratios (commonly >0.5), discordant U–Pb ages and steep MREE-HREE patterns with negative Eu anomalies. Recrystallization would decrease REE and other trace element concentrations and reallocate them among adjoining minerals, a process that could have resulted in the lower REE and trace element concentrations such as P, Ti, Hf, Nb, U, Th and Y in zircon bright domains than cores of samples YC20 and 13YCB (Chen et al., 2010). Under solid state recrystallization conditions, oxygen diffusivity in zircon is sluggish (Zheng et al., 2004), and therefore recrystallized domains have the ability to retain primary $\delta^{18}O$ signatures of magmatic zircons as shown by the similar oxygen isotope compositions in the bright domains and cores of samples YC20 and 13YCB (Zheng et al., 2004). The recrystallized bright domains probably have experienced metamictization when radiogenic lead would be preferentially expelled from lattice in view of the fact that Pb^{4+} has a larger ion radius and thus is less stable than Zr^{4+}, resulting in radiogenic lead loss and the formation of discordant ages in the bright domains of samples YC20 and 13YCB (Vavra et al., 1996). The recrystallized domains, however, still preserve magmatic characteristics as evidenced by its high Th/U ratios (>0.5). As such, recrystallization on zircon bright domains resulted in different yet inherent mineral chemistry and isotopic compositions from precursor magmatic zircons. The 690–680 Ma ages acquired from the bright zircon bright domains represent the ages of recrystallization useless for constraining the depositional age of sampling horizon. The reason why some zircon grains bear no bright domains may be because their rims may have not experienced metamictization, and thus no recrystallization occurred.

5.2. Depositional ages and stratigraphic correlation

Lithological combined with climatic stratigraphic and regional tectonic correlations prompted some geologists to deem the Liantuo Formation as the middle-lower Fulu Formation (Lin, 1995; Xue et al., 2001; Peng et al., 2004; Lin et al., 2013), but this is devoid of radiometric age constraints. Lu et al. (1985) suggested the shallow-water Liantuo Formation correlates with the deep-water Chang'an Formation, they being different lithofacies deposited simultaneously. A SHRIMP U–Pb zircon age of 778.4 ± 5.2 Ma obtained from a tuffaceous siltstone bed within the basal Chang'an Formation seem to have supported this correlation scheme (Gao et al., 2013a,b; Yin and Gao, 2013). However, zircon grains producing this age are rounded to varying extent showing a long prismatic morphology, typical of zircons formed in intrusive rocks widely present in South China. As a result, this age suffers from inheritance and is meaningless for constraining the depositional age of sampling horizon.

Based on regional tectonic, paleomagnetic and previous radiometric dates, Huang et al. (1996a,b) suggested the Liantuo Formation could have been deposited simultaneously with the

Xieshuihe Formation, they both postdating the Banxi Group. This correlation scheme subsequently was seemingly supported by the similar siliciclastic composition and palaeodepositional environment between the Liantuo and Xieshuihe formations (Peng et al., 2004), and the differential deformation imparted in the sandstones of Xieshuihe Formation and its underlying Zhangjiawan Formation (Zhang, 1986a,b; Wang and Suo, 1987). Increasing geochronological evidence seems to have also supported the correlation of Liantuo Formation with Xieshuihe Formation. Yin et al. (2003) documented a SHRIMP zircon U–Pb age of 758 ± 23 Ma from a tuffaceous siltstone bed within the upper Xieshuihe Formation. This age seem to be within uncertainty indistinguishable from the SHRIMP zircon U–Pb age of 748 ± 12 Ma from a tuffaceous siltstone bed within the upper Liantuo Formation (Ma et al., 1984). However, the ca. 758 Ma age documented from the Xieshuihe Formation suffer from inheritance and it is the only one highly concordant U–Pb age of 687 ± 13 Ma acquired from one euhedral zircon grain in Yin et al.'s (2003) paper mostly approximates the true depositional age of Xieshuihe Formation, indicating its possible correlation with the middle-lower Fulu Formation (HBGMR, 1997; Zhang et al., 2008b).

To summarize, neither the Chang'an Formation nor the Fulu Formation nor the Xieshuihe Formation can be correlated with the Liantuo Formation. Only the Banxi Group can be correlative with the Liantuo Formation, both of which were deposited following the breakup of Yangtze Rodinia and predating the initiation of Neoproterozoic glaciations, both contain similar microflora fossil assemblage, and both are dominated by purple sandstone–siltstone composition (Liu et al., 1991; Liu and Li, 1992; Liu et al., 1999). Our newly obtained radiometric dates from the Liantuo Formation have also supported this correlation scheme. Field observations suggest an erosive contact between the Liantuo Formation and its overlying Nantuo Formation in the Three Gorge areas, indicating the true top Liantuo Formation has been eroded off, albeit to an unquantifiable extent. Assuming minor erosion has occurred on the 'top' (the quotation marks used here mean the top Liantuo Formation now is not its original top because of erosion/truncation) Liantuo Formation, the Liantuo Formation would have been deposited at ca. 780–714 Ma. Previous zircon U–Pb ages from intrusions or tuffaceous siltstone beds within the Banxi Group constrained its depositional age to be ca. 814–716 Ma (Gan et al., 1993; Ge et al., 2001; Wang et al., 2003, 2010, 2012a; Yin et al., 2003; Zhang et al., 2008a,d; Gao et al., 2012, 2013b; Lan et al., 2014), which support the correlation of Liantuo Formation with Banxi Group. It should be noted that in a rift basin setting, multiple deposition–erosion–redeposition cycles could occur frequently. As such, zircon grains bearing bright domains of 690–680 Ma from samples YC20 and 13YCB could suffer from inheritance, i.e. they could have been transported from older sedimentary successions. The presence of detrital quartz, feldspar, and mica grains in these samples (Fig. 5e and f) suggest an erosion–transportation–deposition process is not impossible. If so, depositional age for the sampling horizons of samples YC20 and 13YCB should postdate 690–680 Ma, thus the Liantuo Formation should have been deposited in a prolonged period than the Banxi Group. This possibility requires finding more euhedral magmatic zircon grains of 690–680 Ma from Liantuo or other formations in the Nanhua Basin. Another scenario is that appreciable amount of erosion has occurred on the 'top' of the Liantuo Formation. In this regard, the ca. 714 Ma age would not represent the true depositional age of the 'top' Liantuo Formation. Rather, deposition of the topmost Liantuo Formation should be linked to glacial erosion and glaciotectonic activities. The Nantuo glaciers would advance over the loosely bound sediments of the Liantuo Formation, causing erosion and frontal deformation (c.f. Ingólfsson and Lokrantz, 2003). During this process, some tuffaceous materials and other sediments of the Liantuo Formation and/or Nantuo formations were transported upwards and entrained in the ice by means of glacial erosion. Following melting of proceeding Nantuo ice sheets, pebbles were released and deposited simultaneously with the surrounding marine sediments. The presence of thin diamictite on the top Liantuo Formation, despite a thickness of 30–150 cm, would be the lithological evidence for such a process. In this case, depositional age of the upper Liantuo Formation should be <730 Ma. Certainly, this explanation requires discovery of more lines of field geological evidence such as sigma structures isoclinal folded and shear structures (c.f. Ingólfsson and Lokrantz, 2003).

5.3. Matching with paleogeographic evolution

That the Liantuo Formation correlates with Banxi Group is also feasible in terms of regional stratigraphic correlation and paleogeographic evolution. Prior to 820 Ma, the Yangtze Block collided with the Cathaysia Block along the Jiangnan Orogen to form South China Block. During this period, the Sibao Group and their stratigraphic equivalents were deposited around the Hunan–Guizhou–Guangxi area (Wang et al., 2012b). The Huangling Granite intruded around the Three Gorges area creating a high topography and inducing a series of deformation structures (Ji et al., 2014). During 820–790 Ma, rifting occurred along the Jiangnan Orogen from south to north and, correspondingly, accommodated deep and shallow water sediments, respectively as evidenced by the mudstone-dominated compositions in the lower Banxi Group around the Hunan–Guizhou–Guangxi areas and pebbly sandstone in the lower Banxi Group around the Shimen area, Hunan Province (Fig. 2). The Liantuo Formation was situated at a paleohigh position because of intrusion by Huangling Granite and thus received no sedimentation. After ca. 780 Ma, the study area increasingly became inundated as a result of increasing transgression from south to north, forming estuarine and delta depositional settings where deposited a suit of pebbly sandstone. During 780–715 Ma, the Hubei–Hunan–Guizhou–Guangxi areas continue to accommodate sedimentation and deposit dominant siliciclastic sediments in the upper Liantuo Formation and upper Banxi Group. During 720–715 Ma, the Sturtian glaciation occurred with ice probably covered the whole ocean surface, lowering the sea level and bringing a cold and dry climate as evidenced by geochemical proxies (Feng et al., 2006; Zhao et al., 2011). Sedimentation continued but with much slower rates in both continental and oceanic depositional settings.

Following the retreat of Sturtian glaciation (715–690 Ma), diamictite of the Chang'an Formation were deposited but only around western Hunan, Guizhou and Guangxi areas (Fig. 2). No diamictite was present around the Three Gorges and Shimen, Hunan areas, probably because of erosion induced by uplifting and tectonotopography (e.g. Zhang et al., 2008d). During 690–685 Ma (our unpublished age data), continued sedimentation around the western Hunan, Guizhou and Guangxi areas formed the Fulu Formation (Liangjiehe Member and Gucheng Member). Increased transgression because of melting of ice sheets prompted the Shimen, Hunan area to evolve an estuarine-delta depositional setting where deposited pebbly sandstone and sandstone–mudstone of the Xieshuihe Formation. Meanwhile, the study area remained in a paleohigh position continuing to subject to erosion. During the deposition of the upper Xieshuihe Formation, the Gucheng glaciation occurred with ice probably covered the whole ocean surface, lowering the sea level and bringing a cold and dry climate as evidenced by geochemical proxies (Feng et al., 2003, 2004). Following the retreat of Gucheng glaciation, diamictite of the Gucheng Member and then shale of the Datangpo Formation were deposited around the southern Hubei–Hunan–Guizhou–Guangxi areas. However, the study area was still situated at a paleohigh position

accommodating no sedimentation. Sedimentation in this area only began following the retreat of Nantuo glaciation.

5.4. Implications for the initiation of Sturtian glaciation

The ca. 716 Ma from the upper Banxi Group (Lan et al., 2014) and the ca. 714 Ma age from the 'top' Liantuo Formation suggest the Jiangkou glaciation initiated at ca. 715 Ma. Bowring et al. (2007) obtained an ID-TIMS U–Pb zircon age of 713.7 ± 0.5 Ma from the Gubrah Formation interpreted to represent the most approximate onset age of the Sturtian glaciation in Oman (Bowring et al., 2007). Nonetheless, the sampling horizon is above the base of the diamictite, indicating the Sturtian glaciation in Oman should have initiated prior to ca. 714 Ma. Additionally, Macdonald et al. (2010) reported an ID-TIMS U–Pb zircon age of 716.5 ± 0.2 Ma from a tuff bed within the diamictite of the Upper Mount Harper Group in northwestern Canada, which is within errors indistinguishable from the ca. 716–714 Ma ages documented from the 'top' Liantuo Formation and upper Banxi Group. As such, the newly obtained radiometric ages from South China suggest the Sturtian glaciation occurred synchronously ca. 715 Ma in at least three paleocontinents: Arabia-Nubia, Laurentia, and South China.

5.5. Implication for the evolution of Rodinia supercontinent

A long term dispute existed concerning the configuration and breakup history of the Rodinia supercontinent (Li and Evans, 2011; Li et al., 2013a,b). Obtaining more reliable radiometric age controlled paleomagnetic data in Neoproterozoic stratigraphic records from individual continent becomes an urgent solution to resolve this debate. Our newly obtained ca. 714 Ma age from the 'top' Liantuo Formation is much younger than previously dated and, moreover, the stratigraphic estimation of 714–635 Ma for the Liantuo and Nantuo formations is in accord with the initial breakup of Rodinia defined between 720–630 Ma (Li et al., 2013a,b), which, therefore, seem to have coordinated the age-paleopole contradictions to some extent.

6. Conclusions

Zircon grains from six tuff/tuffaceous siltstone beds within the Liantuo Formation gave four reasonable ages of ca. 780–770 Ma, ca. 735–730 Ma, ca. 714 Ma, 690–680 Ma, respectively. The ca. 780–770 Ma zircons are sampled from tuff beds within the basal Liantuo Formation and thus suggest the initial deposition of Liantuo Formation at ca. 780 Ma. The ca. 735–730 Ma zircons are collected from tuffaceous siltstone beds within the middle-upper Liantuo Formation, thereby indicating a ca. 735–730 Ma depositional age for the middle-upper Liantuo Formation. The ca. 714 Ma zircons are sampled from tuffaceous siltstone beds within the 'top' Liantuo Formation, thus indicating a ca. 714 Ma depositional age for the 'top' Liantuo Formation. New radiometric dates suggest the correlation of Liantuo Formation with Banxi Group providing less erosion has occurred on the 'top' Liantuo Formation. Moreover, new ages not only coordinated the age-paleopole contradictions in the Liantuo and Nantuo formations for exploring the evolution history of Rodinia supercontinent, but also strengthened the viewpoint that the Sturtian glaciation occurred at ca. 720 Ma in at least three paleocontinents of Arabia-Nubia, Laurentia, and South China. Oxygen isotopic coupled with REE and trace element compositions suggest that the bright domains of zircon grains in samples YC20 and 13YCB were possibly formed by recrystallization. Hence the 690–680 Ma ages derived from these bright domains represent the age of recrystallization useless for constraining the depositional age of the Liantuo Formation.

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References

Bowring, S.A., Grotzinger, J.P., Condon, D.J., Ramezani, J., Newall, M.J., Allen, P.A., 2007. Geochronologic constraints on the chronostratigraphic framework of the Neoproterozoic Huqf Supergroup Sultanate of Oman. Am. J. Sci. 307, 1097–1145.

Chen, R.X., Zheng, Y.F., Xie, L.W., 2010. Metamorphic growth and recrystallization of zircon: distinction by simultaneous in-situ analyses of trace elements, U–Th–Pb and Lu–Hf isotopes in zircons from eclogite-facies rocks in the Sulu orogen. Lithos 114, 132–154.

Condon, D., Zhu, M.Y., Bowring, S., Wang, W., Yang, A.H., Jin, Y.G., 2005. U–Pb ages from the Neoproterozoic Doushantuo Formation, China. Science 308, 95–98.

Corfu, F., Hanchar, J.M., Hoskin, P.W.O., Kinny, P., 2003. Atlas of zircon textures. Rev. Mineral. Geochem. 53, 469–495.

Du, Q.D., Wang, Z.J., Wang, J., Qiu, Y.S., Jiang, X.S., Deng, Q., Yang, F., 2013. Geochronology and paleoenvironment of the pre-Sturtian glacial strata: evidence from the Liantuo Formatin in the Nanhua rift basin of the Yangtze Block, South China. Precambrian Res. 233, 118–131.

Evans, D.A.D., Li, Z.X., Kirschvink, J.L., Wingate, M.T.D., 2000. A high-quality mid-Neoproterozoic paleomagnetic pole from South China, with implications for ice ages and the breakup configuration of Rodinia. Precambrian Res. 100, 313–334.

Feng, L.J., Chu, X.L., Zhang, Q.R., Zhang, T.G., 2003. CIA (Chemical index of Alteration) and its applications in the Neoproterozoic clastic rocks. Earth Science Frontiers 10, 539–543.

Feng, L.J., Chu, X.L., Zhang, Q.R., Zhang, T.G., Li, H., Jiang, N.F., 2004. New evidence of deposition under cold climate for the Xieshuihe Formation of the Nanhua System in northeastern Hunan China. Chinese Science Bulletin 49, 1420–1427.

Feng, L.J., Chu, X.L., Zhang, T.G., Huang, J., 2006. Liantuo sandstones: sedimentary records under cold climate before the Nanhua big glaciations. Acta Petrol. Sin. 22, 2387–2393 (in Chinese with English abstract).

Gan, X.C., Zhao, F.J., Tang, X.S., 1993. Single Grains of Zircon U–Pb Isoline Age. Evolution of Crust and Mantle and Diagenetic Mineralizational Isotopic Geochemistry. Seismological Press, Beijing, pp. 10–11.

Gao, L.Z., Ding, X.Z., Zhang, C.H., Chen, J., Liu, Y.R., Zhang, H., Liu, X.X., Pang, W.H., 2012. Revised chronostratigraphic framework of the metamorphic strata in the Jiangnan Orogenic Belt, South China and its tectonic implications. Acta Geol. Sin. 86, 339–349.

Gao, L.Z., Ding, X.Z., Yin, C.Y., Zhang, C.H., Ettensohn, F.R., 2013a. Qingbaikouan and Cryogenian in South China: constraints by SHRIMP zircon U–Pb dating. Acta Geol. Sin. 87, 1540–1553.

Gao, L.Z., Lu, J.P., Ding, X.Z., Wang, H.R., Liu, Y.X., Li, J., 2013b. Zircon U–Pb dating of Neoproterozoic tuff in South Guangxi and its implications for stratigraphic correlation. Geol. China 40, 1443–1452 (in Chinese with English abstract).

Gao, W., Zhang, C.H., 2009. Zircon SHRIMP U–Pb ages of the Huangling granite and the tuff beds from Liantuo Formation in the Three Gorges area of Yangtze River, China and its geological significance. Geol. Bull. China 28, 45–50 (in Chinese with English abstract).

Ge, W.C., Li, X.H., Li, Z.X., Zhou, H.W., 2001. Mafic intrusions in Longsheng area: age and its geological implications. Chin. J. Geol. 36, 112–118.

Hoffman, P.F., Li, Z.X., 2009. A palaeogeographic context for Neoproterozoic glaciation. Palaeogeogr. Palaeoclimatol. Palaeoecol. 277, 158–172.

Hoskin, P.W.O., 2005. Trace-element composition of hydrothermal zircon and the alteration of Hadean zircon from the Jack Hills, Australia. Geochim. Cosmochim. Acta 69, 637–648.

Huang, J.Z., Tang, X.S., Guo, L.Q., Zhang, X.Y., 1996a. Paleomagnetic characteristics of Precambrian strata in Hunan and their geological significance. Reg. Geol. China 3, 204–212.

Huang, J.Z., Tang, X.S., Zhang, X.Y., Guo, L.Q., 1996b. On the Correlation between the Liantuo Formation of the Yangtze Gorge and the Banxi Group of Hunan. J. Stratigr. 20, 232–236.

Hunan Bureau of Geology and Mineral Resources, 1997. Stratigraphy (Lithostratic) of Hunan Province. China University of Geosciences, Wuhan, pp. 1–292.

Ingólfsson, Ó., Lokrantz, H., 2003. Massive ground ice body of glacial origin at Yugorski Peninsula, Arctic Russia. Permafr. Periglac. Process. 14, 199–215.

Ji, W.B., Lin, W., Faure, M., Chu, Y., Wu, L., Wang, F., Wang, J., 2014. Origin and tectonic significance of the Huangling massif within the Yangtze craton, South China. J. Asian Earth Sci. 86, 59–75.

Lan, Z.W., Chen, Z.Q., 2012. New xenotime ages obtained from the Paleoproterozoic Kimberley Group, NW Australia: implications for regional hydrothermal events. Aust. J. Earth Sci. 59, 119–133.

Lan, Z.W., Li, X.H., Zhu, M.Y., Chen, Z.Q., Zhang, Q.R., Li, Q.L., Lu, D.B., Liu, Y., Tang, G.Q., 2014. A rapid and synchronous initiation of the Sturtian glaciations. Precambrian Res. 255, 401–411.

Lee, S.G., 1924. A suggestion of a new method for geological survey of igneous intrusions. Acta Geol. Sin. 2, 109–115 (in Chinese with English abstract).

Li, Q.L., Li, X.H., Liu, Y., Tang, G.Q., Yang, J.H., Zhu, W.G., 2010a. Precise U–Pb and Pb–Pb dating of Phanerozoic baddeleyite by SIMS with oxygen flooding technique. J. Anal. At. Spectrom. 25, 1107–1113.

Li, X.H., Li, W.X., Li, Z.X., Liu, Y., 2008a. 850–790 Ma bimodal volcanic and intrusive rocks in northern Zhejiang, South China: a major episode of continental rift magmatism during the breakup of Rodinia. Lithos 102, 341–357.

Li, X.H., Li, W.X., Li, Z.X., Lo, C.H., Wang, J., Ye, M.F., Yang, Y.H., 2009a. Amalgamation between the Yangtze and Cathaysia Blocks in South China: constraints from SHRIMP U–Pb zircon ages, geochemistry and Nd–Hf isotopes of the Shuangxiwu volcanic rocks. Precambrian Res. 174, 117–128.

Li, X.H., Liu, Y., Li, Q.L., Guo, C.H., Chamberlain, K.R., 2009b. Precise determination of Phanerozoic zircon Pb/Pb age by multicollector SIMS without external standardization. Geochem. Geophys. Geosyst. 10, 1–21.

Li, X.H., Li, W.X., Li, Q.L., Wang, X.C., Liu, Y., Yang, Y.H., 2010b. Petrogenesis and tectonic significance of the ~850 Ma Gangbian alkaline complex in South China: evidence from in situ zircon U–Pb dating, Hf–O isotopes and whole-rock geochemistry. Lithos 114, 1–15.

Li, X.H., Tang, G.Q., Gong, B., Yang, Y.H., Hou, K.J., Hu, Z.C., Li, Q.L., Liu, Y., Li, W.X., 2013a. Qinghu zircon: a working reference for microbeam analysis of U–Pb age and Hf and O isotopes. Chin. Sci. Bull. 58, 4647–4654 (in Chinese with English abstract).

Li, W.X., Li, X.H., Li, Z.X., Lou, F.S., 2008b. Obduction-type granites within the NE Jiangxi Ophiolite: implications for the final amalgamation between the Yangtze and Cathaysia blocks. Gondwana Res. 13, 288–301.

Li, Z.X., Wartho, J.A., Occhipinti, S., Zhang, C.L., Li, X.H., Wang, J., Bao, C.M., 2007. Early history of the eastern Sibao Orogen (South China) during the assembly of Rodinia: new mica $^{40}Ar/^{39}Ar$ dating and SHRIMP U–Pb detrital zircon provenance constraints. Precambrian Res. 159, 79–94.

Li, Z.X., Bogdanova, S.V., Collins, A.S., Davidson, A., De Waele, B., Ernst, R.E., Fitzsimons, I.C.W., Fuck, R.A., Gladkochub, D.P., Jacobs, J., Karlstrom, K.E., Lu, S., Natapov, L.M., Pease, V., Pisarevsky, S.A., Thrane, K., Vernikovsky, V., 2008c. Assembly, configuration, and break-up history of Rodinia: a synthesis. Precambrian Res. 160, 179–210.

Li, Z.X., Evans, D.A.D., 2011. Late Neoproterozoic 40 intraplate rotation within Australia allows for a tighter-fitting and longer-lasting Rodinia. Geology 39, 39–42.

Li, Z.X., Evans, D.A.D., Halverson, G.P., 2013b. Neoproterozoic glaciations in a revised global palaeogeography from the breakup of Rodinia to the assembly of Gondwanaland. Sediment. Geel. 294, 219–232.

Lin, S.J., 1995. Correlation for Liantuo Formation to Banxi Group and Boundary between the Presinian and the Sinian in Eastern Guizhou. Guizhou Geol. 12, 23–29 (in Chinese with English abstract).

Lin, S.J., Lu, D.B., Xiao, J.F., Xiong, X.H., Li, Y.T., 2013. Stratigraphy of the Nanhuan System in Guizhou Province. J. Stratigr. 37, 542–557.

Liu, H.Y., Sha, Q.A., 1963. New view of the Sinian System of the Xiedong area along the Changjiang River. Sci. Geol. Sin. 4, 177–187 (in Chinese with English abstract).

Liu, H.Y., Dong, R.S., Lao, Q.Y., Chen, M.E., Zhang, Q.R., Li, J.L., Hu, W.H., Qi, Z.L., 1991. The Sinian System of China. Science Press, Beijing, pp. 388 (in Chinese).

Liu, H.Y., Li, Y.J., 1992. On the age and regional correlation of Banxi Group. Sci. Geol. Sin. Supp., 1–16 (in Chinese with English abstract).

Liu, H.Y., Hao, J., Li, Y.J., 1999. Late Precambrian Stratigraphy in East-central China and Implications for Geological Evolution. Science Press, Beijing, pp. 145–169 (in Chinese).

Liu, W.X., Zhang, B.C., 1989. About the comparison of Wuqiangxi Formation in Hunan and Liantuo Formation in Xiadong District. Hunan Geol. 8, 66–70 (in Chinese with English abstract).

Liu, X.M., Gao, S., Diwu, C.R., Ling, W.L., 2008. Precambrian crustal growth of Yangtze Craton as revealed by detrital zircon studies. Am. J. Sci. 308, 421–468.

Liu, Y., Li, X.H., Li, Q.L., Tang, G.Q., Yin, Q.Z., 2011. Precise U–Pb zircon dating at <5 micron scale by Cameca IMS 1280 SIMS using Gaussian illumination probe. J. Anal. At. Spectrom. 26, 845–851.

Lu, S.N., Ma, G.G., Gao, Z.J., Lin, W.X., 1985. Primary research on glacigenic rocks of the Late Precambrian in China. In: Precambrian Geology Editorial Committee (Ed.), Precambrian Geology, Vol. 1. The Collected Works of Late Precambrian Glacigenous Rocks in China. Geological Publishing House, Beijing, pp. 1–86 (in Chinese with English abstract).

Ludwig, K.R., 2008. Isoplot 3.70: a geochronological toolkit for Microsoft Excel. Berkeley Geochronology Center, Special Publication 4, 77 pp.

Ma, G.G., Li, H.Q., Zhang, Z.C., 1984. An investigation of the age limits of the Sinian System in South China. Bull. Yichang Inst. Geol. Miner. Resour.: Chin. Acad. Geol. Sci. 8, 1–29 (in Chinese).

Macdonald, F.A., Schmitz, M.D., Crowley, J.L., Roots, C.F., Jones, D.S., Maloof, A.C., Strauss, J.V., Cohen, P.A., Johnston, D.T., Schrag, D.P., 2010. Calibrating the Cryogenian. Science 327, 1241–1243.

McCormack, K.D., Mary Gee, M.A., McNaughton, N.J., Smith, R., Fletcher, I.R., 2009. U–Pb dating of magmatic and xenocryst zircons from Mangakino ignimbrites and their correlation with detrital zircons from the Torlesse metasediments, Taupo Volcanic Zone, New Zealand. J. Volcanol. Geotherm. Res. 183, 97–111.

Nasdala, L., Hofmeister, W., Norberg, N., Mattinson, J.M., Corfu, F., Dörr, W., Kamo, S.L., Kennedy, A.K., Kronz, A., Reiners, P.W., Frei, D., Kosler, J., Wan, Y.S., Götze, J., Häger, T., Kröner, A., Valley, J.W., 2008. Zircon M257 – a homogeneous natural reference material for the ion microprobe U–Pb analysis of zircon. Geostand. Geoanal. Res. 32, 247–265.

Pearce, N.J., Perkins, W.T., Westgate, J.A., Gorton, M.P., Jackson, S.E., Neal, C.R., Chenery, S.P., 1997. A compilation of new and published major and trace element data for NIST SRM 610 and NIST SRM 612 glass reference materials. Geostand. Newsl. 21, 115–144.

Peng, X.J., Liu, Y.R., Wu, N.J., Chen, J.C., Li, J.Q., 2004. Correlation of the Nanhuan strata on the southern margin of the Yangtze landmass. J. Stratigr. 28, 354–359 (in Chinese with English abstract).

Sláma, J., Košler, J., Condon, D.J., Crowley, J.L., Gerdes, A., Hanchar, J.M., Horstwood, M.S.A., Morris, G.A., Nasdala, L., Norberg, N., Schaltegger, U., Schoene, B., Tubrett, M.N., Whitehouse, M.J., 2008. Plešovice zircon – a new natural reference material for U–Pb and Hf isotopic microanalysis. Chem. Geol. 249, 1–35.

Stacey, J.S., Kramers, J.D., 1975. Approximation of terrestrial lead isotope evolution by a two-stage model. Earth Planet. Sci. Lett. 26, 207–221.

Steiger, R.H., Jäger, E., 1977. Subcommissions on geochronology: convention on the use of decay constants in geo- and cosmochronology. Earth Planet. Sci. Lett. 36, 359–362.

Tera, F., Wasserburg, G.J., 1972. U–Th–Pb systematics in three Apollo 14 basalts and the problem of initial Pb in lunar rocks. Earth Planet. Sci. Lett. 14, 281–304.

Valley, J.W., Kinny, P.D., Schulze, D.J., Spicuzza, M.J., 1998. Zircon megacrysts from kimberlite: oxygen isotope variability among mantle melts. Contrib. Mineral. Petrol. 133, 1–11.

Vavra, G., Gebauer, D., Schmid, R., Compston, W., 1996. Multiple zircon growth and recrystallization during polyphase Late Carboniferous to Triassic metamorphism in granulites of the Ivrea Zone (Southern Alps): an ion microprobe (SHRIMP) study. Contrib. Mineral. Petrol. 122, 337–358.

Wang, J., Li, Z.X., 2003. History of Neoproterozoic rift basins in South China: implications for Rodinia break-up. Precambrian Res. 122, 141–158.

Wang, J., Li, X.H., Duan, T.Z., 2003. Zircon SHRIMP U–Pb dating for the Cangshuipu volcanic rocks and its implications for the lower boundary age of the Nanhua strata in South China. Chin. Sci. Bull. 48, 1663–1669.

Wang, X.C., Li, X.H., Li, Z.X., Li, Q.L., Tang, G.Q., Gao, Y.Y., Zhang, Q.R., Liu, Y., 2012a. Episodic Precambrian crust growth: evidence from U–Pb ages and Hf–O isotopes of zircon in the Nanhua Basin, central South China. Precambrian Res. 222–223, 386–403.

Wang, Z.J., Wang, J., Duan, T.Z., Xie, Y., Zhuo, J.W., Yang, P., 2010. Geochronology of middle Neoproterozoic volcanic deposits in Yangtze Craton interior of South China and its implications to tectonic settings. Sci. China D: Earth Sci. 53, 1307–1315.

Wang, Z.Q., Suo, S.T., 1987. The middle and upper Proterozoic levels and correlation of Guizhou, Hunan and Jiangxi Provinces. In: Editorial Committee of Precambrian Geology (Ed.), Precambrian Geology 6: The Collected Works of Late Precambrian Geology in China. Geological Publishing House, Beijing, pp. 73–86 (in Chinese).

Wang, Z.Q., Yin, C.Y., Gao, L.Z., Tang, F., Liu, Y.Q., Liu, P.J., 2006a. The character of the chemical index of alteration and discussion of subdivision and correlation of the Nanhua System in Yichang Area. Geol. Rev. 52, 577–585 (in Chinese with English abstract).

Wang, Z.Q., Yin, C.Y., Gao, L.Z., Liu, Y.Q., Tang, F., Zhang, C.H., 2006b. Chemostratigraphic studies to explain Neoproterozoic stratigraphic division and correlation. Earth Sci. Front. 13, 268–279.

Wang, Z.Q., Yin, C.Y., Gao, L.Z., Tang, F., 2009. Chemostratigraphic characteristics of the Nanhua System in southern Guizhou-northern Guangxi area. Acta Geosci. Sin. 30, 465–474.

Wang, Z.Q., Gao, L.Z., Ding, X.Z., Huang, Z.Z., 2012b. Tectonic environment of the metamorphosed basement in the Jiangnan Orogen and its environmental features. Geol. Rev. 58, 401–413 (in Chinese with English abstract).

Wetherill, G.W., 1956. Discordant uranium–lead ages. Trans. Am. Geophys. Union 37, 320–326.

Wiedenbeck, M., Alle, P., Corfu, F., Griffin, W.L., Meier, M., Oberli, F., Vonquadt, A., Roddick, J.C., Speigel, W., 1995. Three natural zircon standards for U–Th–Pb, Lu–Hf, trace-element and REE analyses. Geostand. Newsl. 19, 1–23.

Wu, Y.B., Zheng, Y.F., Tang, J., Gong, B., Zhao, Z.F., Liu, X.M., 2007. Zircon U–Pb dating of water-rock interaction during Neoproterozoic rift magmatism in South China. Chem. Geol. 246, 65–86.

Xue, Y.S., Cao, R.J., Tang, T.F., Yin, L.M., Yu, C.L., 2001. The Sinian stratigraphic sequence of the Yangtze region and correlation to the late Precambrian strata of North China. J. Stratigr. 25, 207–234 (in Chinese with English abstract).

Yin, C.Y., Liu, D.Y., Gao, L.Z., Wang, Z.Q., Xing, Y.S., Jian, P., Shi, Y.R., 2003. Lower boundary age of the Nanhua system and the Gucheng glacial stage: evidence from SHRIMP dating. Chin. Sci. Bull. 48, 1657–1662 (in Chinese with English abstract).

Yin, C.Y., Wang, Y.G., Tang, F., Wan, Y.S., Wang, Z.Q., Gao, L.Z., Xing, Y.S., Liu, P.J., 2006. SHRIMP II U–Pb zircon date from the Nanhuan Datangpo Formation in Songtao County, Guizhou Province. Acta Geol. Sin. 80, 273–278 (in Chinese with English abstract).

Yin, C.Y., Gao, L.Z., 2013. Definition, time limit and stratigraphic subdivision of the Nanhuan System in China. J. Stratigr. 37, 534–541 (in Chinese with English abstract).

Zeck, H.P., Williams, I.S., 2002. Inherited and magmatic zircon from Neogene Hoyazo cordierite dacite, SE Spain-anatectic source rock provenance and magmatic evolution *in memoriam* Professor Chris Powell. J. Petrol. 43, 1089–1104.

Zhang, L.H., 1986a. Sedimentary facies and stratigraphic division of the upper Proterozoic Banxi Group, Xieshuihe Formation in Yangjiaping region, Northwestern Hunan. In: Wang, H.Z., Yang, W.R., Liu, B.P. (Eds.), The Structural Evolution

of Paleo-Continental Margin in South China. China University of Geosciences (Wuhan) Press, pp. 124–135 (in Chinese).
Zhang, Q.R., Piper, J.D.A., 1997. Palaeomagnetic study of Neoproterozoic glacial rocks of the Yangtze Block: palaeolatitude and configuration of South China in the late Proterozoic Supercontinent. Precambrian Res. 85, 173–199.
Zhang, Q.R., Chu, X., Bahlburg, H., Feng, L., Dobrzinski, N., Zhang, T., 2003. Stratigraphic architecture of the Neoproterozoic glacial rocks in the Xiang-Qian-Gui region of the central Yangtze Block, South China. Progr. Natl. Sci. 13, 783–787.
Zhang, Q.R., Chu, X.L., 2006. The stratigraphic classification and correlation of the Jiangkou glaciation in the Yangtze Block and the stratotype section of the Nanhuan System. J. Stratigr. 30, 306–314 (in Chinese with English abstract).
Zhang, Q.R., Li, X.H., Feng, L.J., Huang, J., Song, B., 2008a. A new age constraint on the onset of the Neoproterozoic glaciations in the Yangtze Platform, South China. J. Geol. 116, 423–429.
Zhang, Q.R., Chu, X.L., Feng, L.J., 2008b. A correlation of the Xieshuihe Formation, Nanhuan System, with a discussion to its glacial sedimentary structures. J. Stratigr. 32, 246–252 (in Chinese with English abstract).
Zhang, Q.R., Chu, X.L., Feng, L.J., 2011. Neoproterozoic glacial records in the Yangtze Region, China. In: Arnaud, E., Halverson, G.P., Shields-Zhou, G. (Eds.), The Geological Record of Neoproterozoic Glaciations, vol. 36. Geological Society, London, Memoirs, pp. 357–366.
Zhang, S.H., Jiang, G.Q., Han, Y.G., 2008c. The age of the Nantuo Formation and Nantuo glaciation in South China. Terra Nova 20, 289–294.
Zhang, S.H., Jiang, G.Q., Dong, J., Han, Y.G., Wu, H.C., 2008d. New SHRIMP U–Pb age from the Wuqiangxi Formation of Banxi Group: implications for rifting and stratigraphic erosion associated with the early Cryogenian (Sturtian) glaciation in South China. Sci. China D: Earth Sci. 51, 1537–1544.
Zhang, S.H., Evans, D.A.D., Li, H.Y., Wu, H.C., Jiang, G.Q., Dong, J., Zhao, Q.L., Raub, T.D., Yang, T.S., 2013. Paleomagnetism of the late Cryogenian Nantuo Formation and paleogeographic implications for the South China Block. J. Asian Earth Sci. 72, 164–177.
Zhang, S.H., Li, H.Y., Jiang, G.Q., Evans, D.A.D., Dong, J., Wu, H.C., Yang, T.S., Liu, P.J., Xiao, Q.S., 2014. New paleomagnetic results from the Ediacaran Doushantuo Formation in South China and their paleogeographic implications. Precambrian Res., http://dx.doi.org/10.1016/j.precamres.2014.09.018 (in press).
Zhang, S.L., 1986b. Outline of the tectonic-palaeogeography and sedimentary characteristics in the Late Proterozoic, Northwest Hunan. In: Wang, H.Z., Yang, W.R., Liu, B.P. (Eds.), The Structural Evolution of Paleo-Continental Margin in South China. China University of Geosciences (Wuhan) Press, pp. 113–123 (in Chinese).
Zhao, X.M., Liu, S.D., Zhang, Q.X., Wu, J.H., Zeng, B.F., Liao, Z.M., Yang, G.Z., Li, F.H., 2011. Geochemical characters of the Nanhua System in Changyang, western Hubei Province and its implications for climate and sequence correlation. Acta Geol. Sin. 85, 576–585.
Zheng, Y.F., Wu, Y.B., Chen, F.K., Gong, B., Li, L., Zhao, Z.F., 2004. Zircon U–Pb and oxygen isotope evidence for a large-scale ^{18}O depletion event in igneous rocks during the Neoproterozoic. Geochim. Cosmochim. Acta 68, 4145–4165.
Zhou, C.M., Tucker, R., Xiao, S.H., Peng, Z.X., Yuan, X.L., Chen, Z., 2004. New constraints on the ages of Neoproterozoic glaciations in south China. Geology 32, 437–440.